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GOVERNO DO BRASIL
DO LADO DO POVO BRASILEIRO

GODS

ON THE FORMATiON OF SATELLiTES iN DENSE SOLiD-PARTiCLE DiSKS

GUSTAVO MADEiRA

15TH FRIENDS OF FRIENDS ANNUAL MEETING

- Assistant Researcher at the National Observatory (Brazil)
- PhD in Physics from UNESP (Brazil)
- Postdoctoral researcher at the Institut de Physique du Globe de Paris (France) in cosmochemistry
- Former member of the DART/NASA mission
- Member of Hera/ESA and RAMSES/ESA missions
- P.I. of the Group of Origin and Dynamics of Satellites (GODS)

Research Topics

- Formation of Planetary Satellites
- Orbital Dynamics and Evolution
- Internal Structure and Composition of Satellites
- Formation and Evolution of Planetary Rings
- Evolution of Circumplanetary Disks

On the formation of satellites in dense solid-particle disks

G. Madeira [a,b,*], L. Esteves [c,b], T.F.L.L. Pinheiro [a], P.V.S. Soares [c], N.S. Santos [c], B. Morgado [d]

[a] Observatório Nacional/MCTI, Rio de Janeiro, RJ, 20921-400, Brazil
[b] Université Paris Cité, Institut de Physique du Globe de Paris, CNRS, Paris, F-75005, France
[c] Orbital Dynamics and Planetology Group (GDOP), São Paulo State University-UNESP, Guaratinguetá, SP, 12516-410, Brazil
[d] Universidade Federal do Rio de Janeiro - Observatório do Valongo, Rio de Janeiro, RJ, 20.080-090, Brazil

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ABSTRACT

Single massive satellites are of great observational interest, as they can produce prominent and potentially detectable signatures. For terrestrial planets and super-Earths, giant impacts in the late stages of formation may generate dense self-gravitating disks — favourable environments for the formation of such satellites. Motivated by this, we explore satellite formation in dense solid-particle disks through three-dimensional N-body simulations, focusing on the effects of disk mass and the surface density exponent (β). Our results reveal significant variability in the masses and configurations of satellites formed under identical disk parameters, highlighting the stochastic nature of the process. Higher disk masses and flatter surface density profiles favour the formation of more massive satellites. Disks with masses above 0.03 planetary masses typically yield a single dominant satellite, while those between 0.003 and 0.03 tend to form two-satellite systems. On average, the mass of the largest satellite scales linearly with the initial disk mass, in agreement with analytical predictions. We estimate that a disk with a minimal mass of 0.03 planetary masses around a 1.6 Earth-mass planet orbiting a Sun-like star could form an Earth–Moon-like system detectable by telescopes with a photometric precision of 10 parts per million – a level achievable by the James Webb Space Telescope.

Madeira et al. (2025b)

Do self-gravitating hard-sphere disk simulations form Darwin ellipsoids?

G. Madeira[1,a], L. Esteves[2], and T. F. L. L. Pinheiro[1]

[1] Observatório Nacional/MCTI, Rio de Janeiro 20921-400, Brazil
[2] Orbital Dynamics and Planetology Group (GDOP), São Paulo State University, Guaratinguetá 12516-410, Brazil

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Abstract The equilibrium shape of homogeneous fluid satellites results from a balance between self-gravity, planetary tides, and rotation, corresponding to Darwin ellipsoids—equilibrium figures associated with the Love numbers h_s. In this work, we simulate self-gravitating hard-sphere disks that produce a single Moon-mass satellite. The satellite's shape is tracked over time to assess whether it resembles a Darwin ellipsoid, as predicted by theory. The Moon-like satellite forms near the planet's Roche limit and grows by accreting both disk particles and recently coalesced proto-satellites. The energy dissipated during the merger impacts rapidly drives the satellite toward a spin-orbit equilibrium state and a triaxial ellipsoidal shape. Our results show that the numerical and analytical shapes exhibit discrepancies below 30%, indicating that this class of simulations is capable of reproducing the expected equilibrium shapes reasonably well, provided that the particle resolution of the disk is sufficiently high ($\gtrsim 10^5$ particles). The discrepancies decrease for satellites formed at larger distances from the planet, where weaker gravitational perturbations from both the planet and the remnant disk allow for greater structural homogenization.

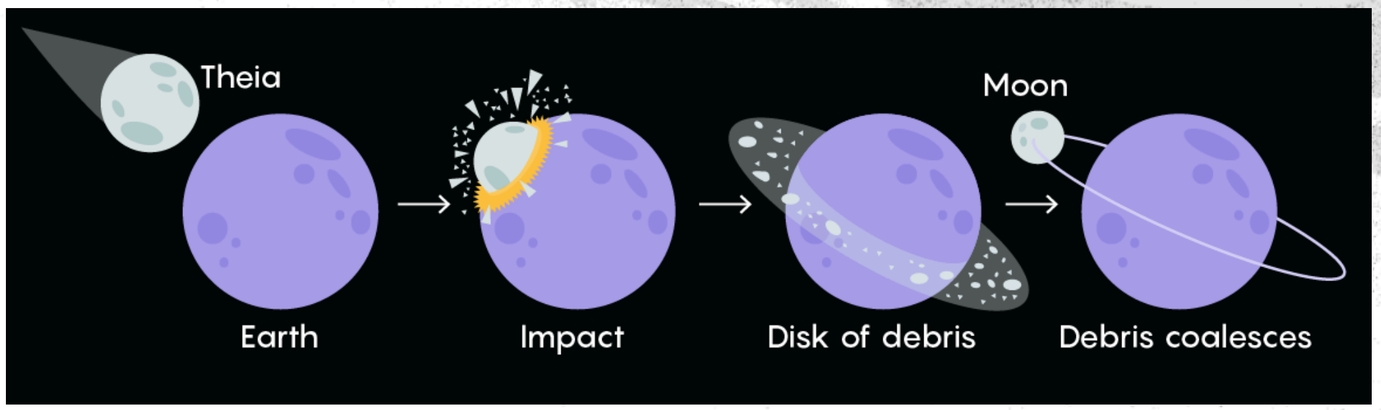
Madeira et al. (2025c)

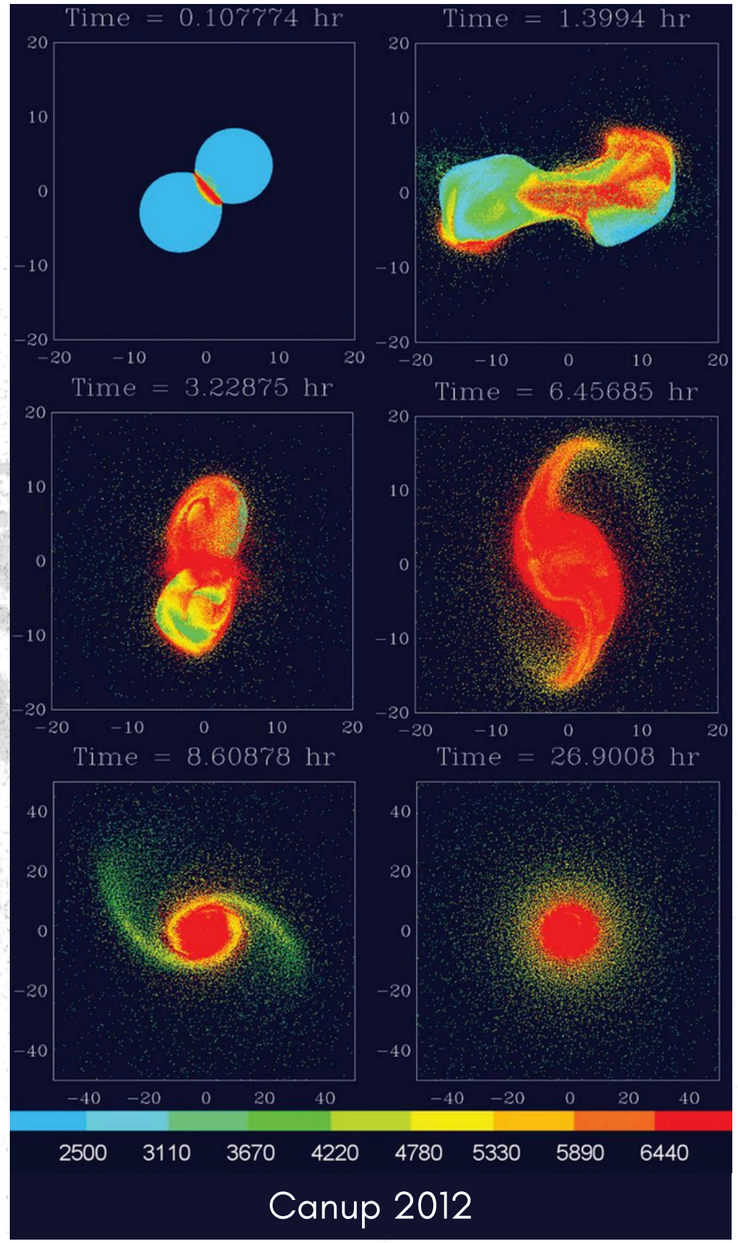
In these works, we explore satellite formation around super-Earths, assuming a process analogous to the formation of the Moon and possibly the Martian moons.

BUT HOW DiD THE SATELLiTES OF THE TERRESTRiAL PLANETS FORM?

- The formation of terrestrial planets is characterized by giant impacts between planetesimals (e.g., Raymond et al., 2006; Izidoro et al., 2021, 2022).

- The satellites of Earth and Mars are thought to have formed from debris disks produced by giant impacts (e.g., Canup, 2012; Lock et al., 2018; Citron et al., 2025; Madeira et al., 2023a).

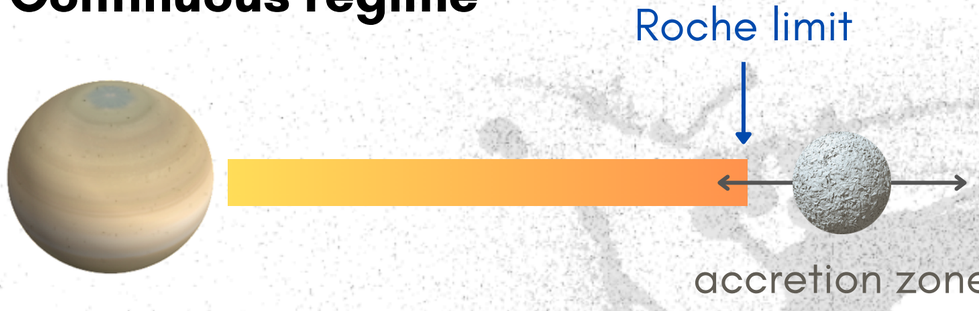




HOW DOES SATELLiTE FORMATiON PROCEED iN A SOLiD DiSK?

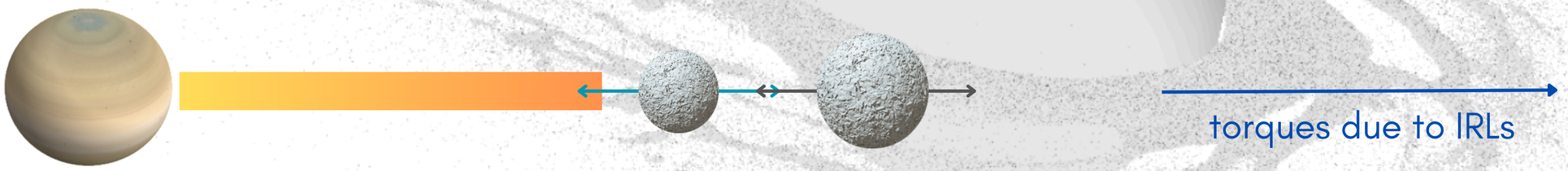
According to Crida & Charnoz (2012), satellite formation in a solid disk proceeds in three distinct stages.

Continuous regime

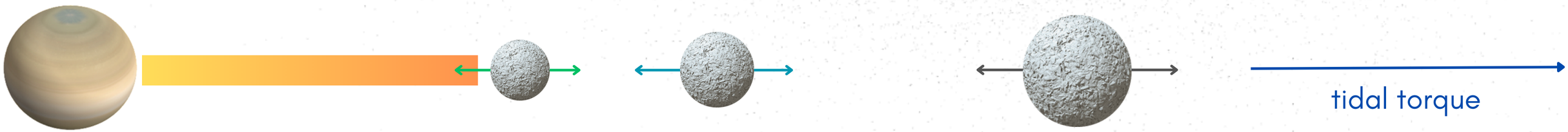


These regimes are derived from one-dimensional hydrodynamical simulations.

Discrete regime



Pyramidal regime



NUMERiCAL SiMULATiONS

- In our simulations, we use two collision treatment schemes:
  - **Pure Hard Sphere** (more accurate but much slower)
  - **Condition accretion scheme** (It loses tidal strippings, but much faster)

- Conditional accretion scheme:

NUMERiCAL SiMULATiONS

- Numerical simulations with a disk of 50,000 particles, randomly distributed from the central body's radius to the Roche limit.

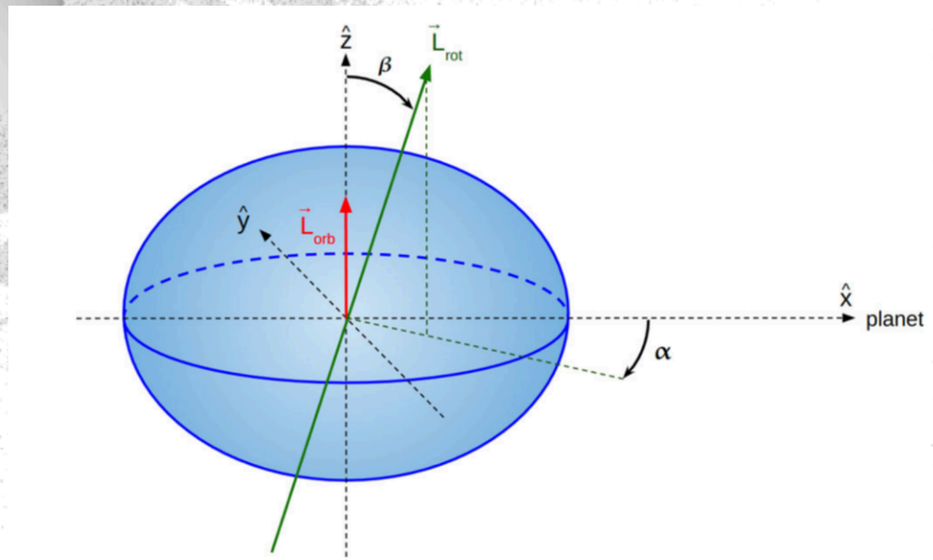
$$\Sigma(a) = \frac{M_{\rm disk}}{\pi(a_{\rm RL}^2 - R_{\rm C}^2)} \left[\int_{R_{\rm C}}^{a_{\rm RL}} \left(\frac{a}{R_{\rm c}} \right)^{-\beta} da \right]^{-1} \left(\frac{a}{R_{\rm c}} \right)^{-\beta}$$

- β = 0, 0.5, 0.75, 3 e 5.
- Mdisk = 0.003, 0.005, 0.008, 0.01, 0.03, 0.05, 0.08 e 0.1 central masses
- 30 simulations for each set of β and Mdisk for the conditional accretion scheme
- The initial angular momentum of the disk is kept constant within each set
- Simulations are run until the disk mass decreases to 5% of its initial value
- For selected cases, we perform simulations using pure hard spheres to track the shape of the formed satellites

SATELLiTE SHAPE

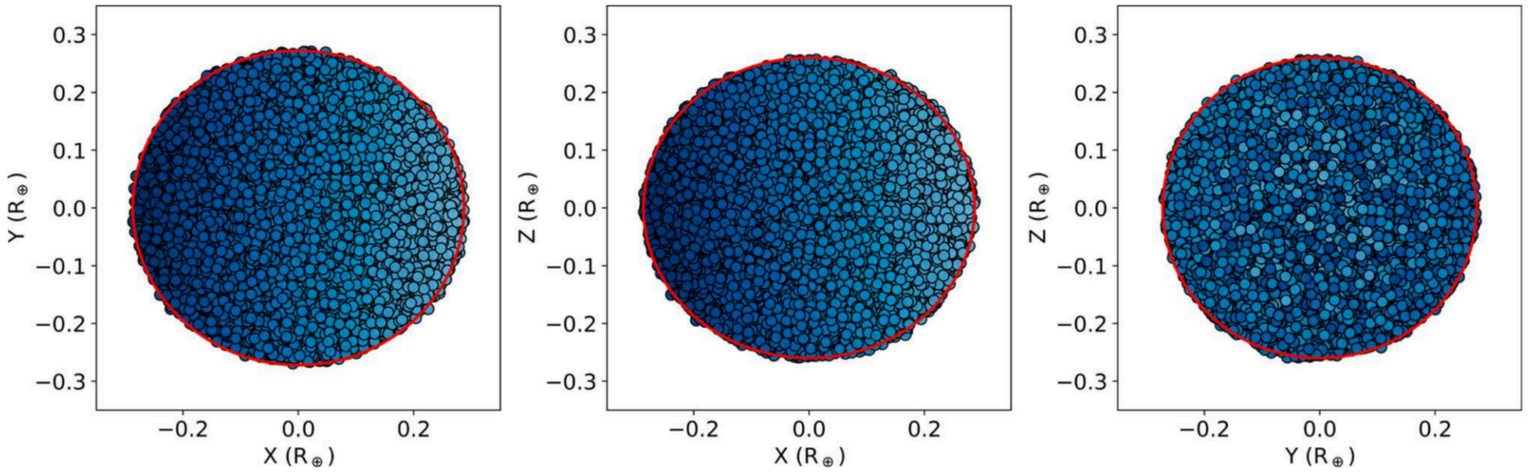
- The shape of a satellite results from the balance between rotation, planetary tides, self-gravity, and material strength
- We track the satellite shape employing two different methods:
  - Numerical procedure: satellite's shape is directly extracted from the simulation data
  - Analytical procedure: estimates the satellite's theoretical equilibrium shape, assuming a perfect, self-gravitating, incompressible fluid satellite

- The shape is computed based on a local orbital coordinate frame (LOC frame):
  - centred on the satellite's centre of mass
  - x-axis points towards the planet
  - z-axis is perpendicular to the orbital plane
  - y-axis completes the right-handed orthonormal basis
- α is the longitude of the spin axis
- β is the colatitude of the spin axis



NUMERiCAL PROCEDURE

- We compute the position of the center of mass, the velocity, and the orbital angular momentum of the satellite in the barycentric reference frame
- These quantities are used to define the LOC frame



Moon-sized satellite composed of 28914 particles

- The position and velocity in the LOC reference frame of the particles that make up the satellite are used to calculate the angular momentum of rotation and the moment of inertia:

$$\mathbf{I} = \begin{pmatrix} \sum m_i(y_i^2 + z_i^2) & -\sum m_i x_i y_i & -\sum m_i x_i z_i \\ -\sum m_i x_i y_i & \sum m_i(x_i^2 + z_i^2) & -\sum m_i y_i z_i \\ -\sum m_i x_i z_i & -\sum m_i y_i z_i & \sum m_i(x_i^2 + y_i^2) \end{pmatrix}$$

- The ellipsoid axes are computed:

$$a = \sqrt{\frac{5}{2M}(I_2 + I_3 - I_1)}, \quad b = \sqrt{\frac{5}{2M}(I_1 + I_3 - I_2)}, \quad c = \sqrt{\frac{5}{2M}(I_1 + I_2 - I_3)}$$

ANALYTiCAL PROCEDURE

- In the LOC frame, a point on the satellite's surface is described by:

$$x_s = r_s\lambda = r_s \sin\theta \cos\phi$$
$$y_s = r_s\mu = r_s \sin\theta \sin\phi$$
$$z_s = r_s\nu = r_s \cos\theta$$

- The equilibrium shape will be given by:

$$r(\theta, \phi) = R_s \left(1 + q \sum_{j=2}^{4} h_j P_j(\lambda) \left(\frac{R_s}{a_s} \right)^{j+1} - \frac{1}{3} h_2 (1+q) F_p^2 \left(\frac{R_s}{a_s} \right)^3 P_2(\cos\Theta) \right)$$

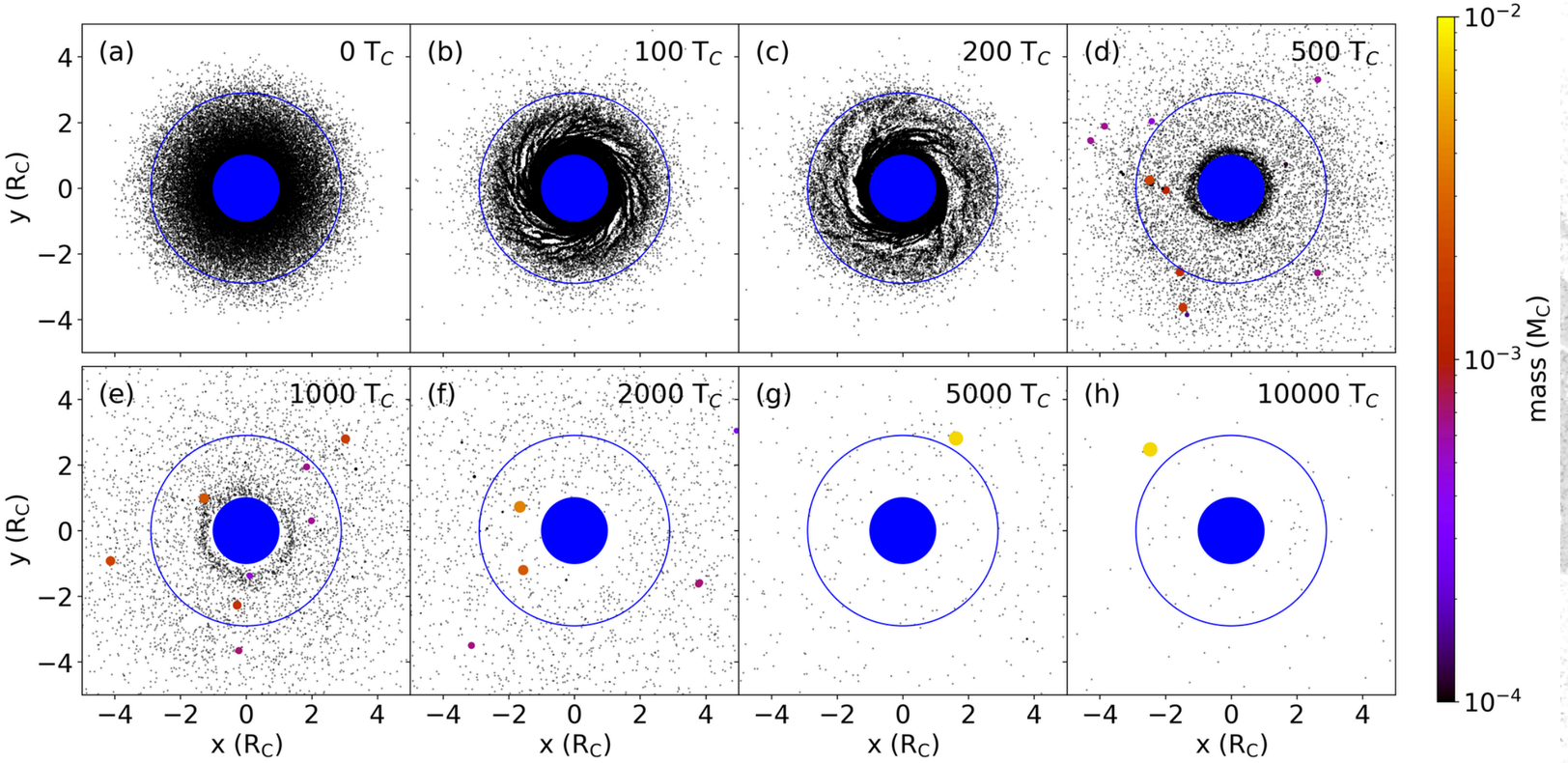
- where

$$h_j = \frac{2j+1}{2(j-1)}, \quad \cos\Theta = \lambda \sin\beta \cos\alpha + \mu \sin\beta \sin\alpha + \nu \cos\beta$$

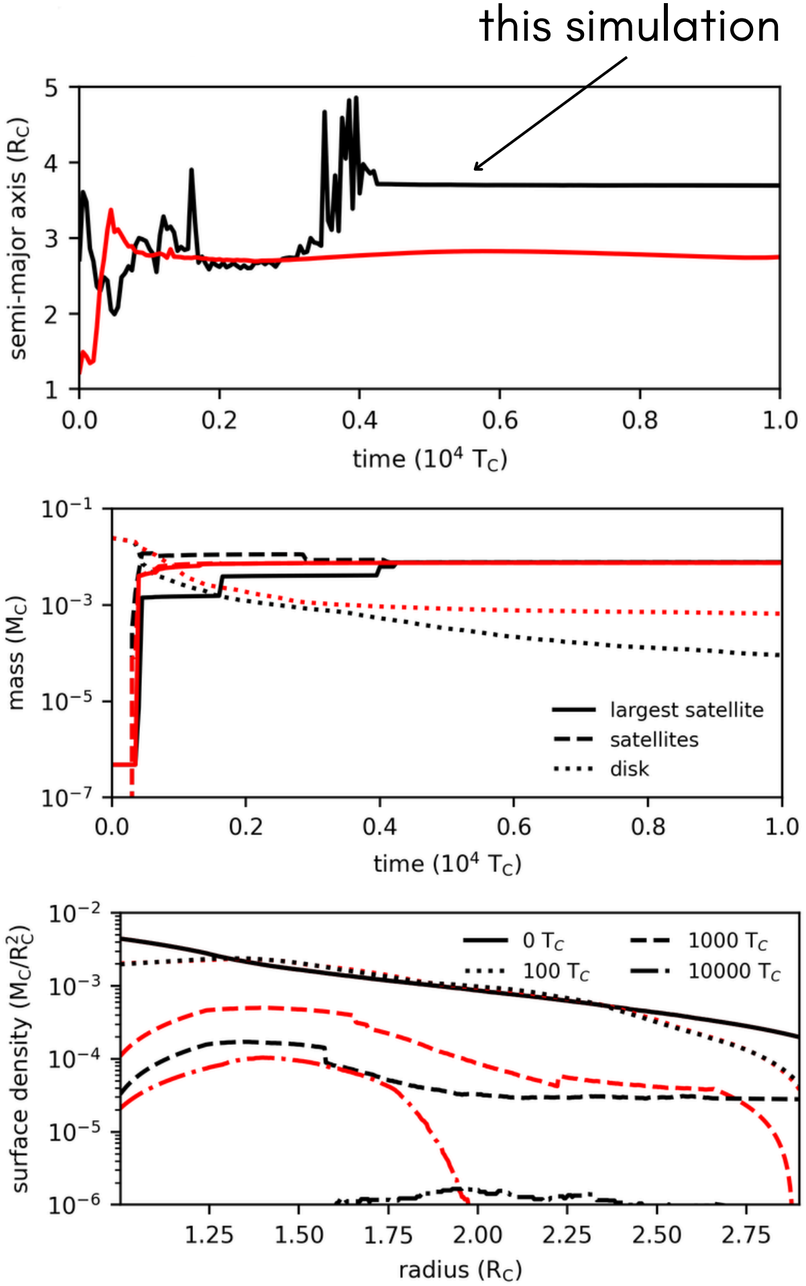
- Semi-axes obtained by evaluating a = r(0°,0°), b = r(0°, 90°) and c=r(90°, 0°)

THE THREE-DiMENSiONAL FORMATiON OF A SATELLiTE

Example of a system with β = 0.75 and Mdisk = 0.03 Mc
(high-mass disk)

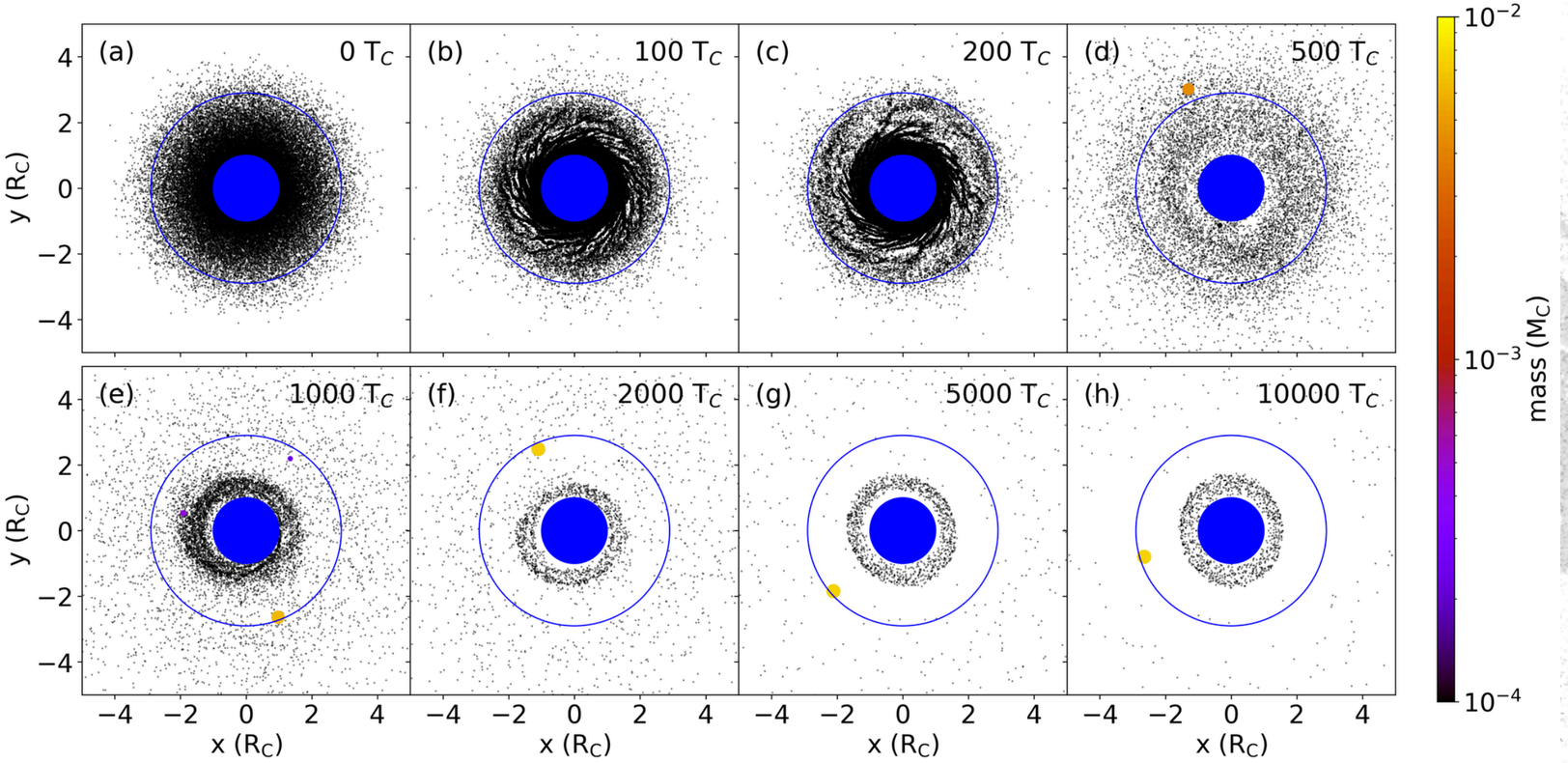


Formation of a lunar-mass satellite with a final eccentricity of 0.15



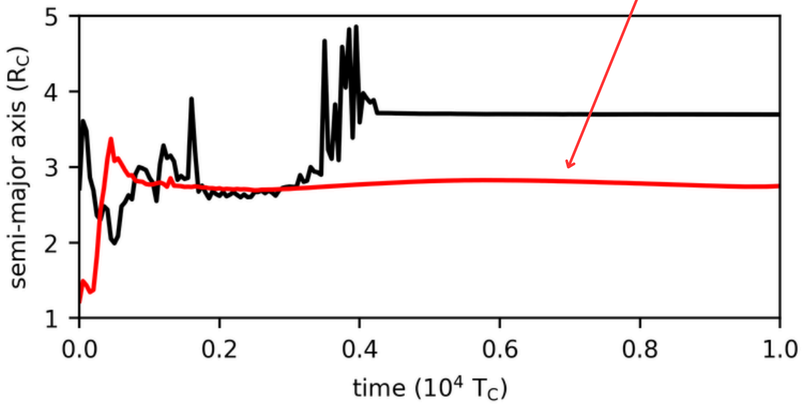
THE THREE-DiMENSiONAL FORMATiON OF ANOTHER SATELLiTE

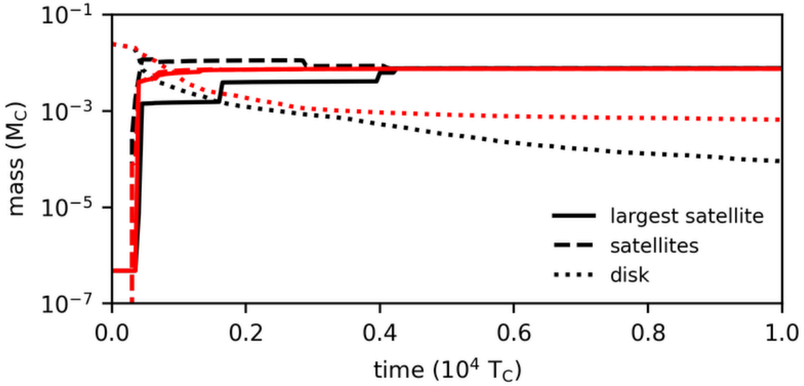
Example of another system with β = 0.75 and Mdisk = 0.03 Mc (high-mass disk)

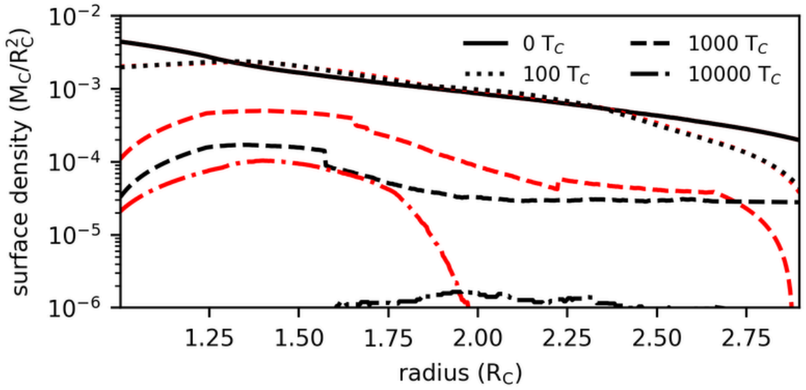


Formation of a lunar-mass satellite with a final eccentricity of 0.01 and a confined ring

this simulation

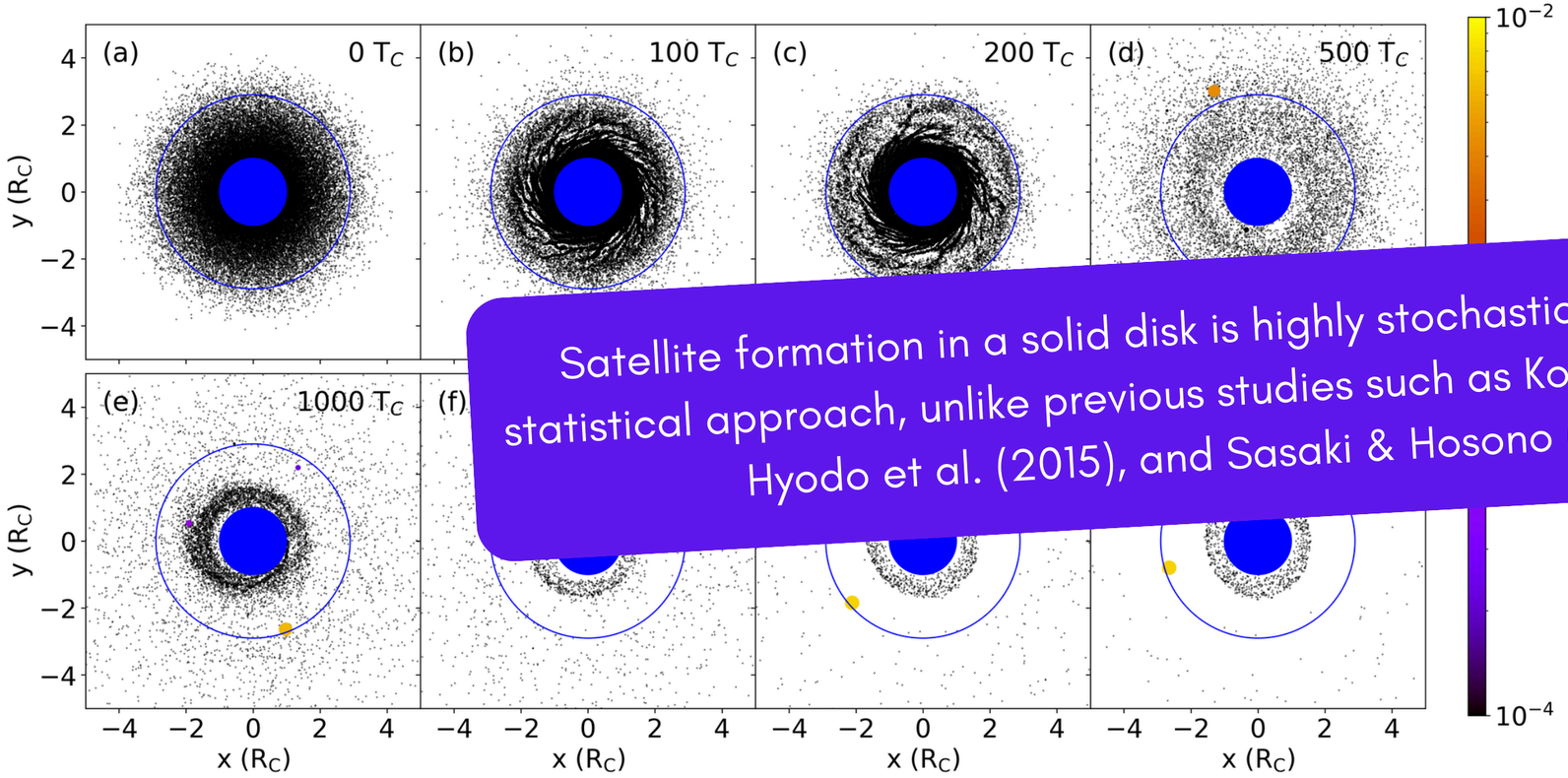




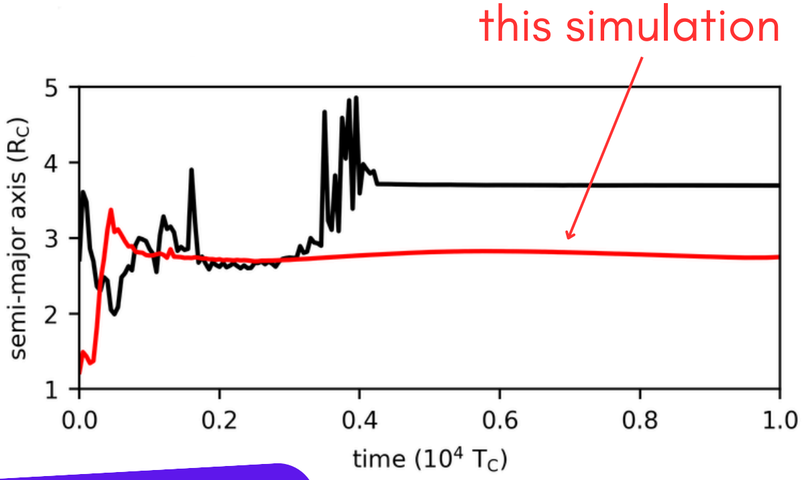


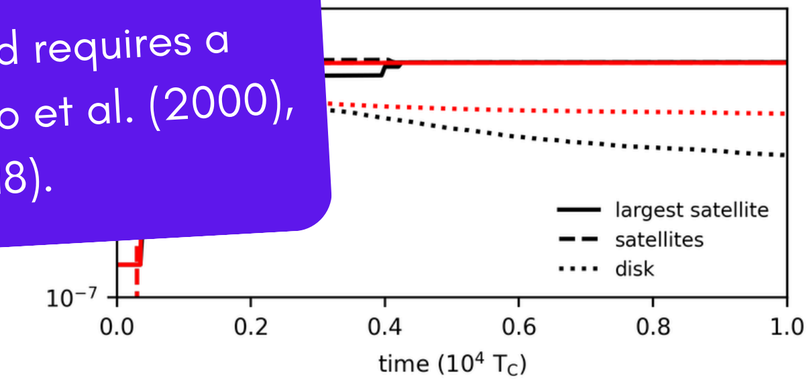
THE THREE-DiMENSiONAL FORMATiON OF ANOTHER SATELLiTE

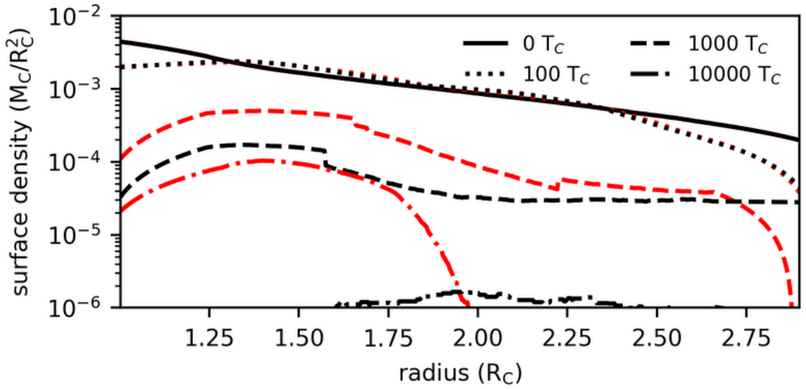
Example of another system with β = 0.75 and Mdisk = 0.03 Mc (high-mass disk)



Satellite formation in a solid disk is highly stochastic and requires a statistical approach, unlike previous studies such as Kokubo et al. (2000), Hyodo et al. (2015), and Sasaki & Hosono (2018).



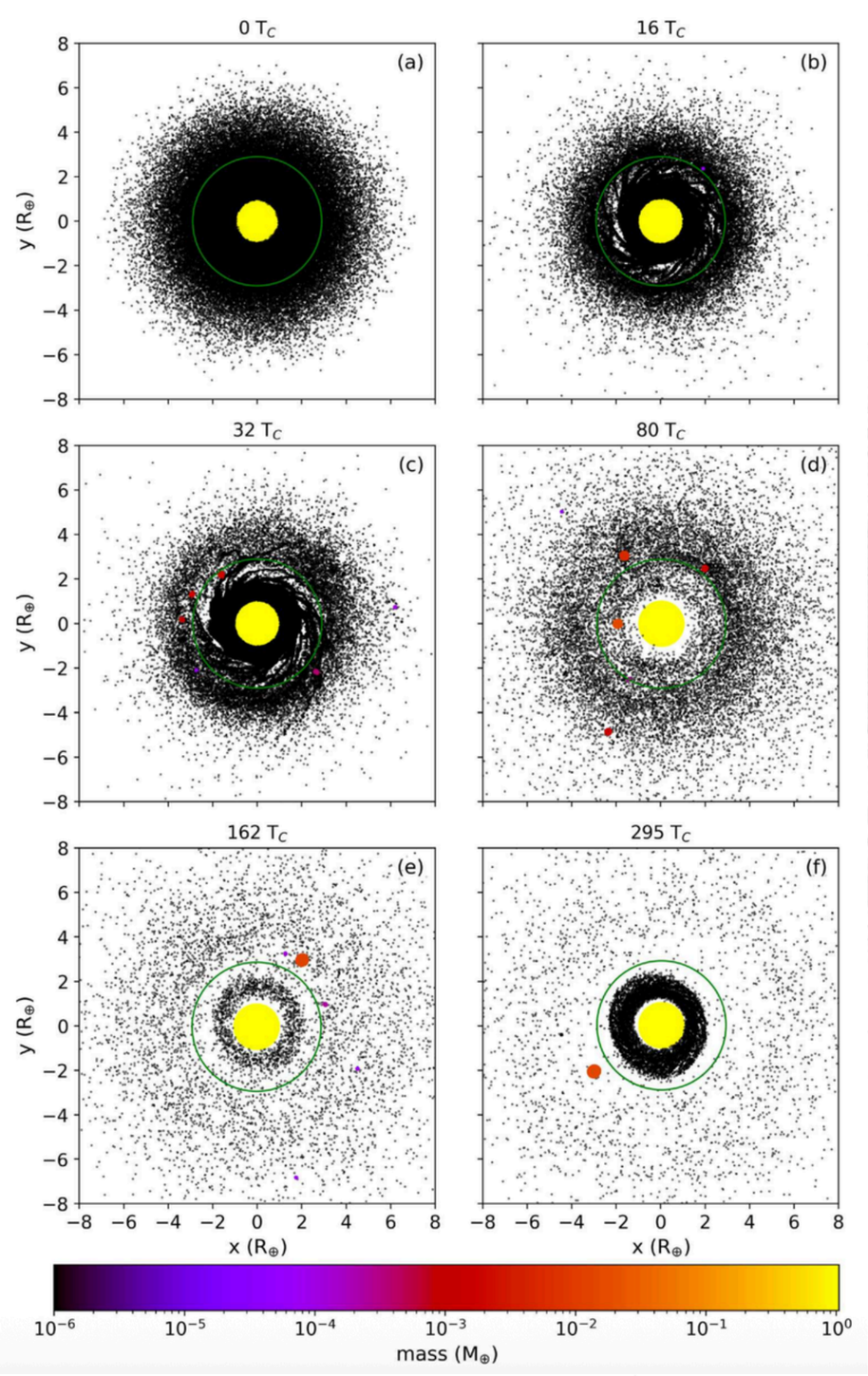


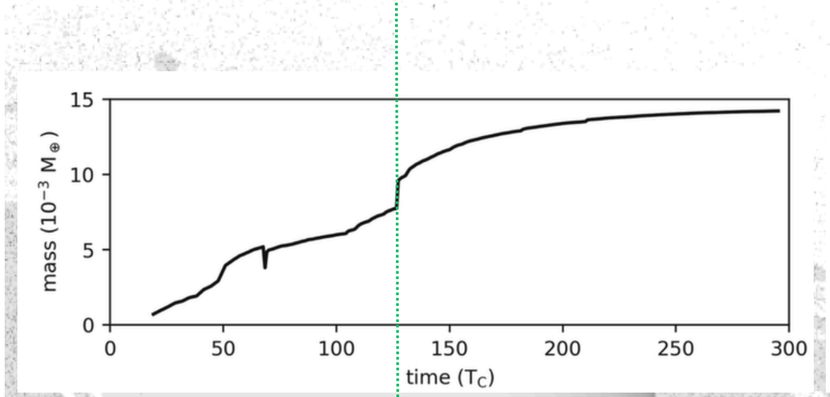


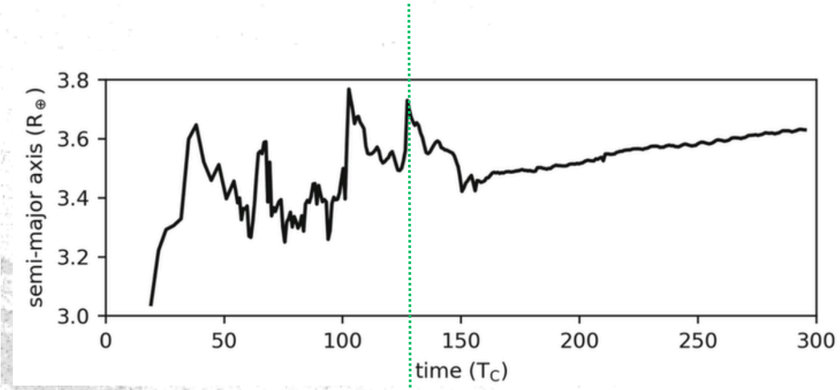
Formation of a lunar-mass satellite with a final eccentricity of 0.01 and a confined ring

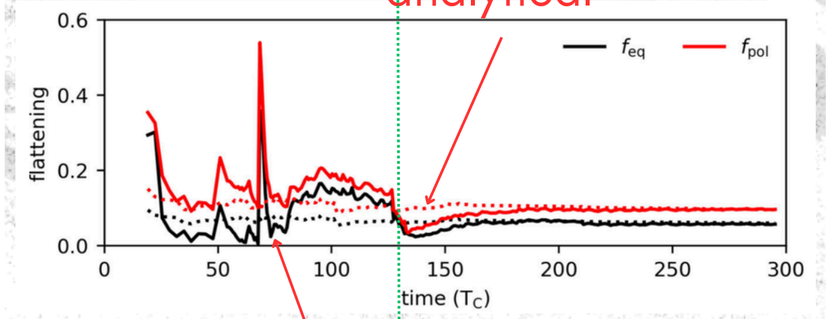
ON THE SHAPE OF THE SATELLiTES

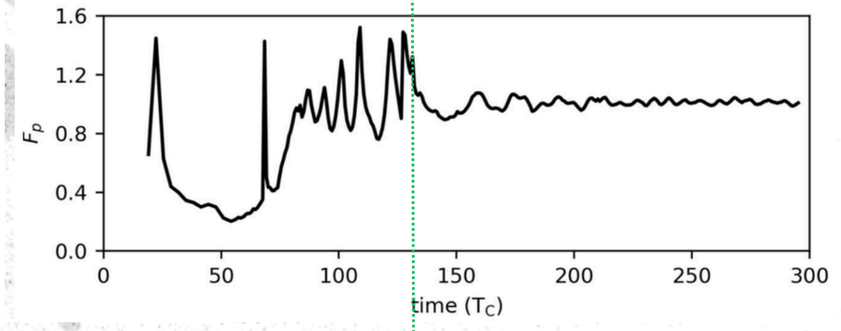
- For this set of parameters, we performed 5 simulations with hard-sphere scheme







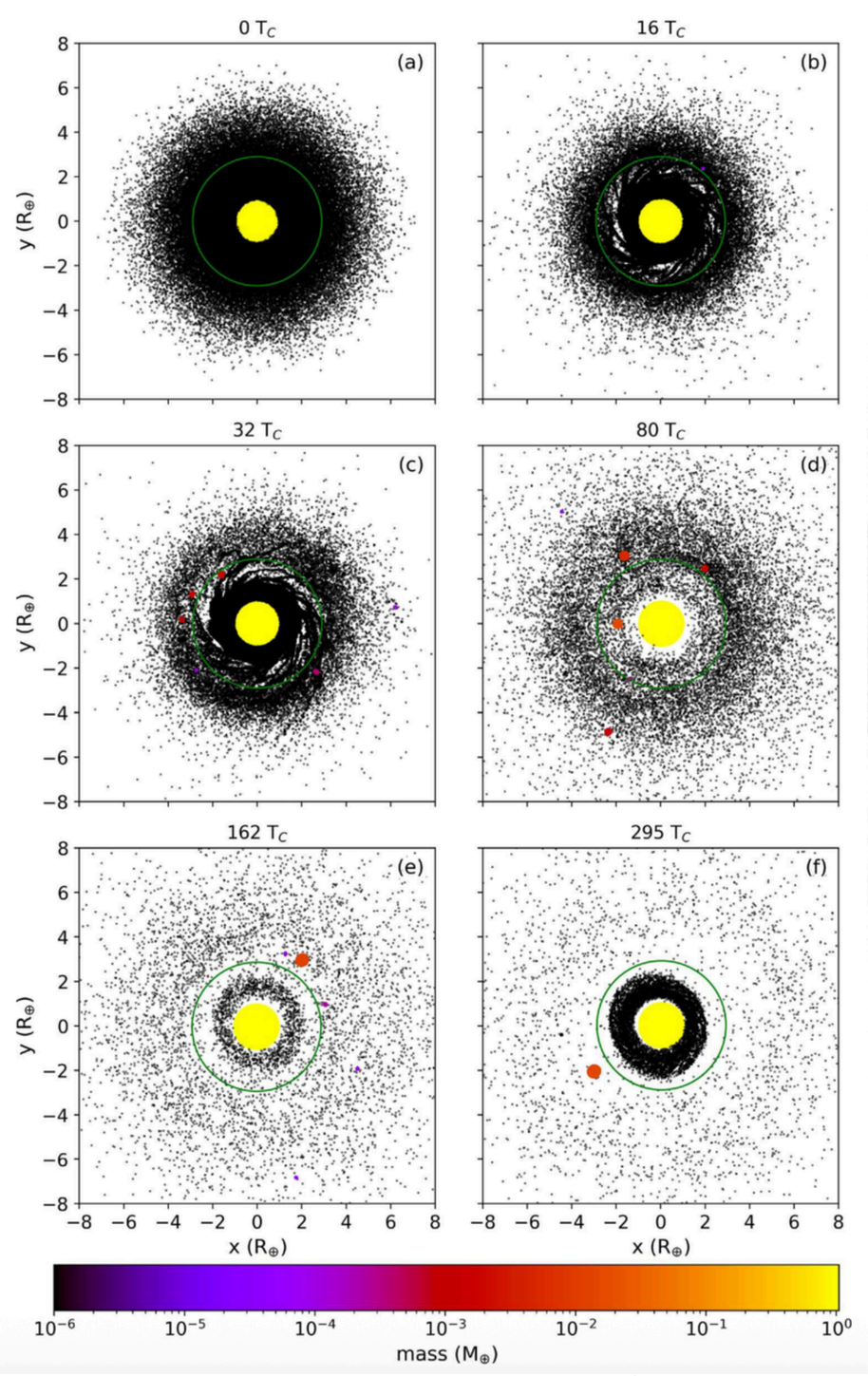


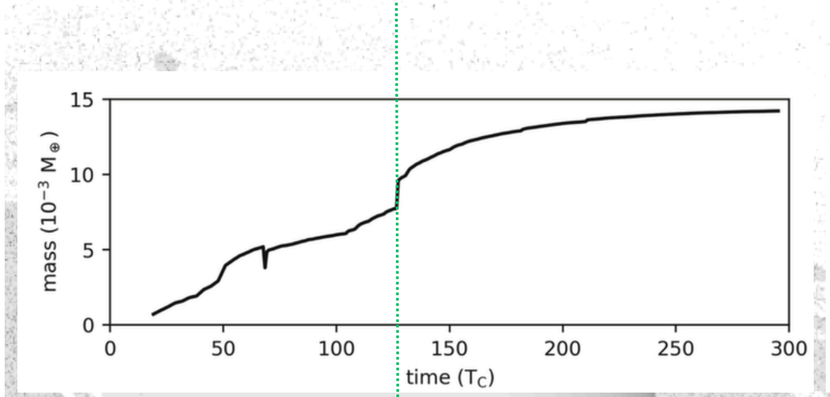


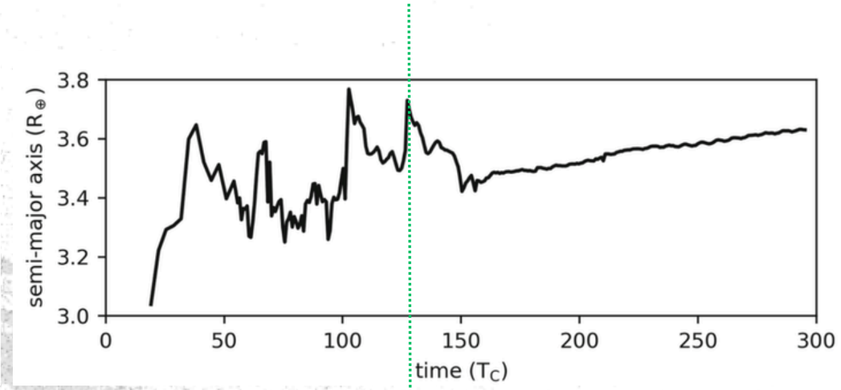
Example case: Formation of a lunar-mass satellite with a final eccentricity of 0.02 and a confined ring

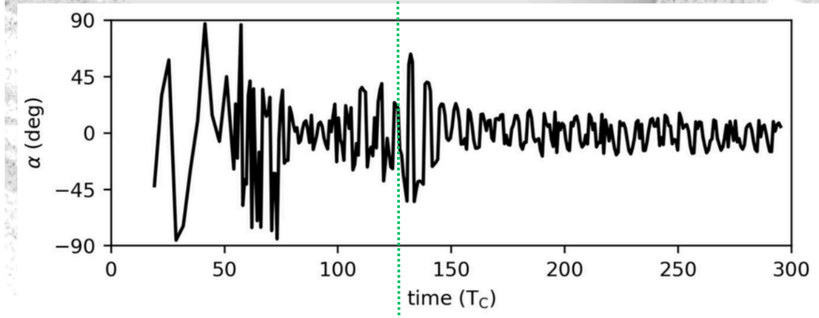
ON THE SHAPE OF THE SATELLiTES

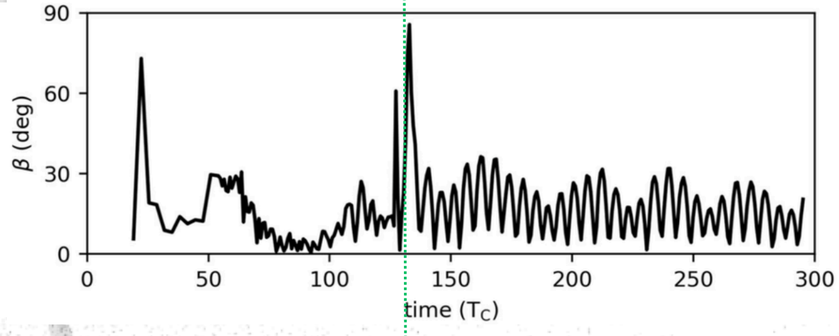
- For this set of parameters, we performed 5 simulations with hard-sphere scheme







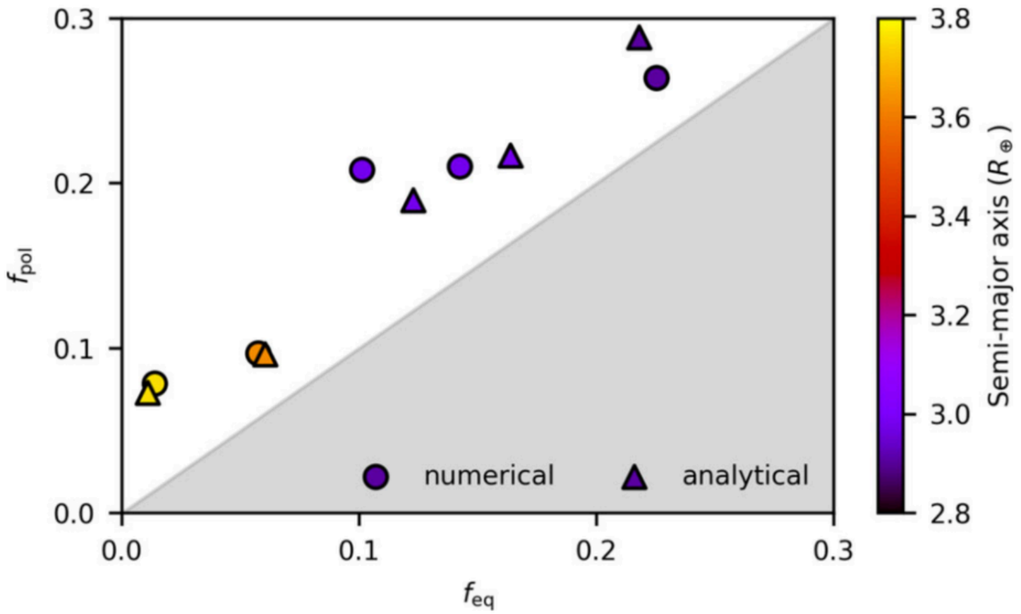




Example case: Formation of a lunar-mass satellite with a final eccentricity of 0.02 and a confined ring

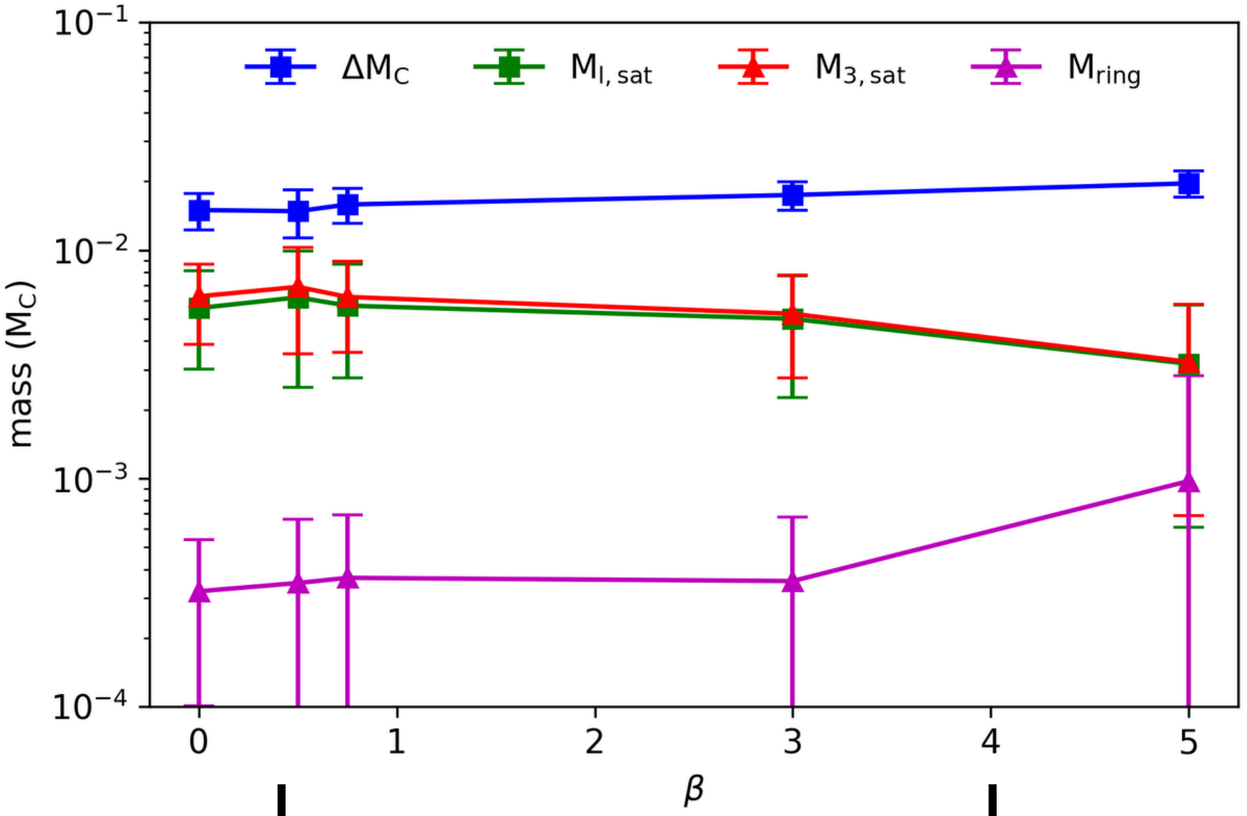
ON THE SHAPE OF THE SATELLiTES

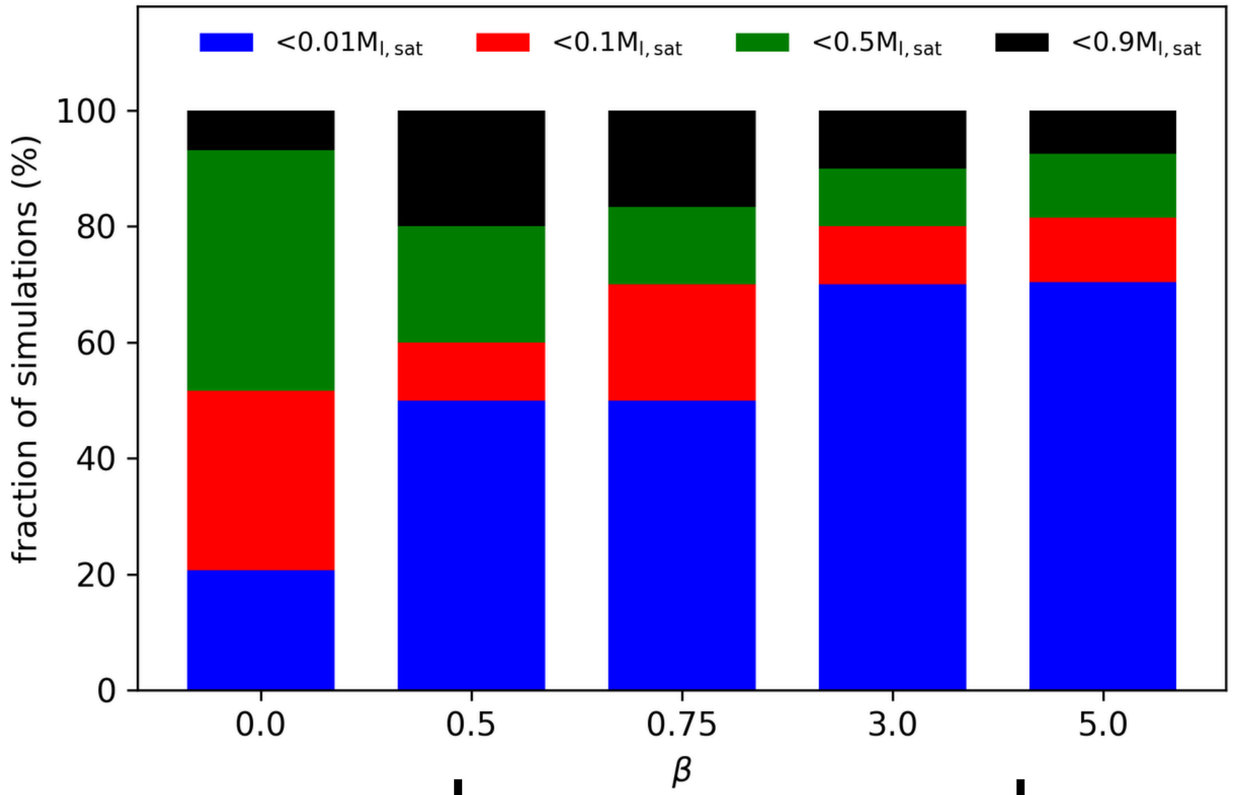
Outcome of the 5 simulations:



- Numerical and analytical shapes exhibit discrepancies below 30%.
- Discrepancies decrease with a larger semi-major axis, while no correlation was found with the satellite's mass.
- Hard-sphere simulations reproduces the expected equilibrium shapes reasonably well, provided that the particle resolution of the disk is sufficiently high (>10,000 particles).
- Discrepancies decrease for satellites formed at larger distances from the planet.
- At larger distance, weaker gravitational perturbations from both the planet and the remnant disk allow for greater structural homogenization.

ON THE DiSK MASS DiSTRiBUTiON





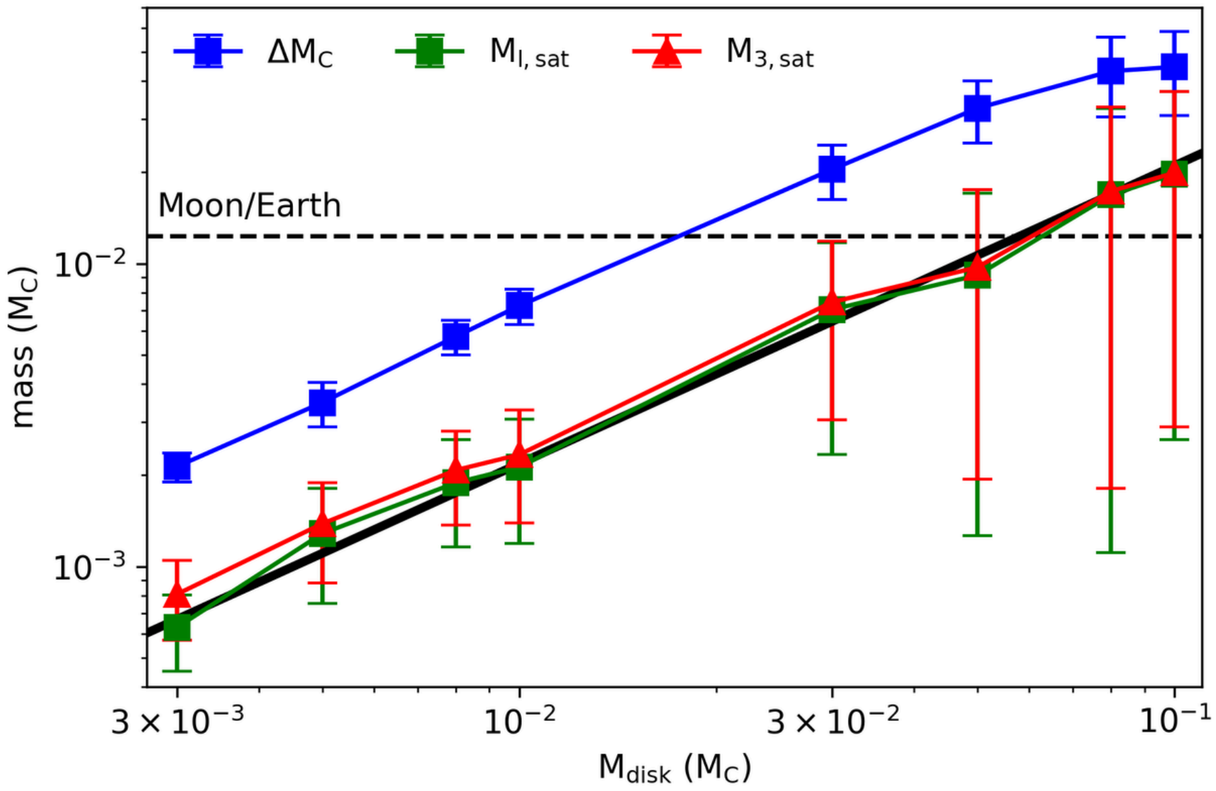
shallower profiles

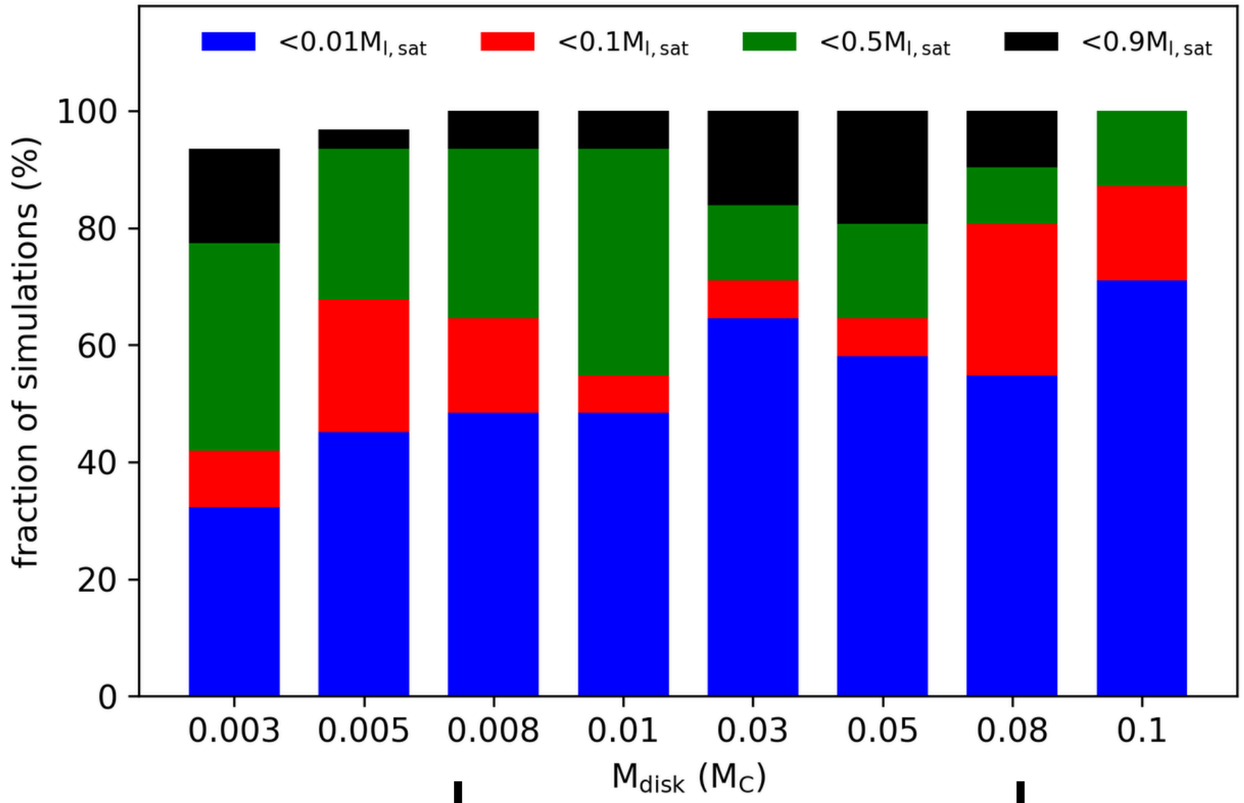
steeper profiles

two dominant satellites

one dominant satellite

ON THE DiSK MASS



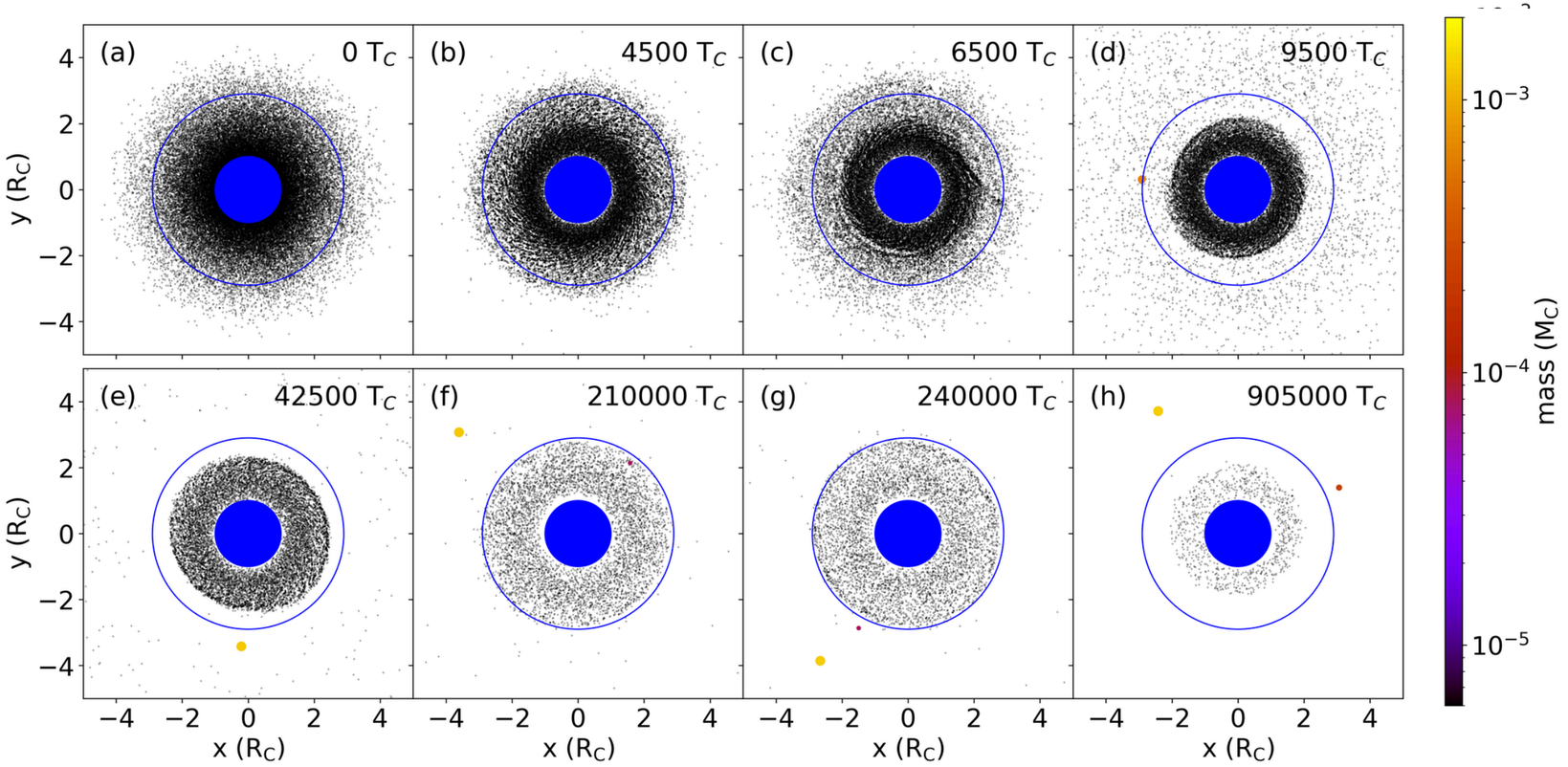


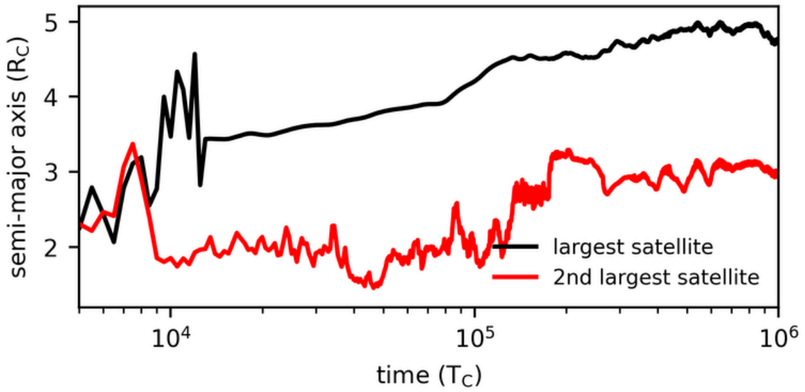
discrete regime

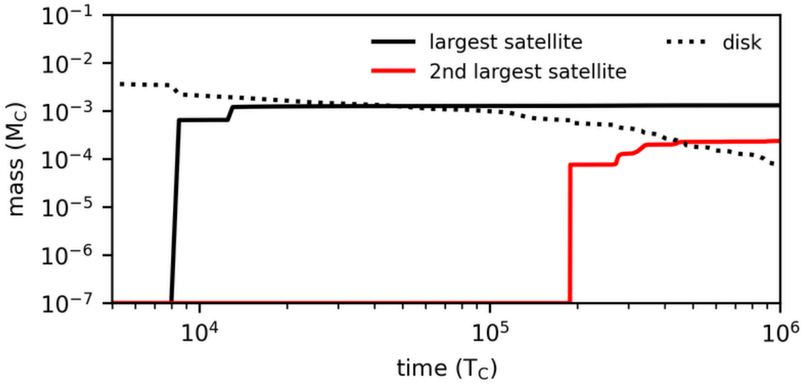
continuous regime

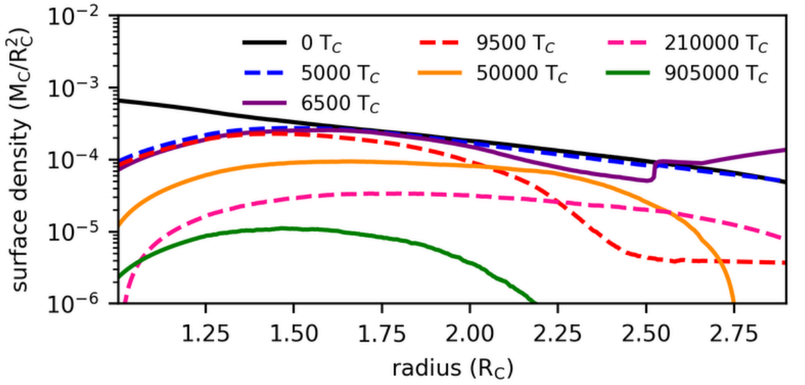
FORMATiON iN THE DiSCRETE REGiME

Example of another system with β = 3 and Mdisk = 0.005 Mc
(intermediate-mass disk)









Formation of two satellites and a confined ring

UNDER WHAT CONDiTiONS DOES THE PYRAMiDAL REGiME OCCUR?

Continuous regime (Mdisk > 0.05 Mc)

zona de acreção

Discrete regime (0.005 Mc ⩽ Mdisk ⩽ 0.05 Mc)

Pyramidal regime (0.005 Mc < Mdisk)

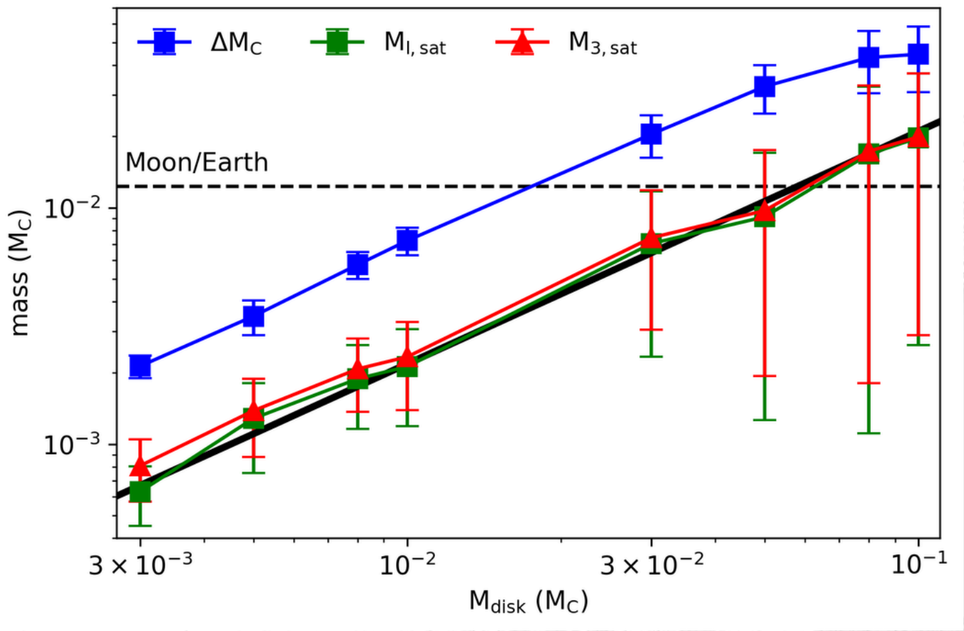
Migration due to planetary tides

$$t_{\text{tide}} = \left(\frac{k_2}{Q}\right)^{-1} \frac{a^5}{3nM_{1,\text{sat}}},$$

- For super-Earths and terrestrial planets:
  - k_2/Q = 0.001–0.01
  - Mdisk = 0.003 Mc
  - ttide = 10^6–10^7 orbital periods
  - tform = 10^5 orbital periods
- On the pyramidal regime: Madeira et al. (2023a,b), Madeira & Charnoz (2024)

tidal torque

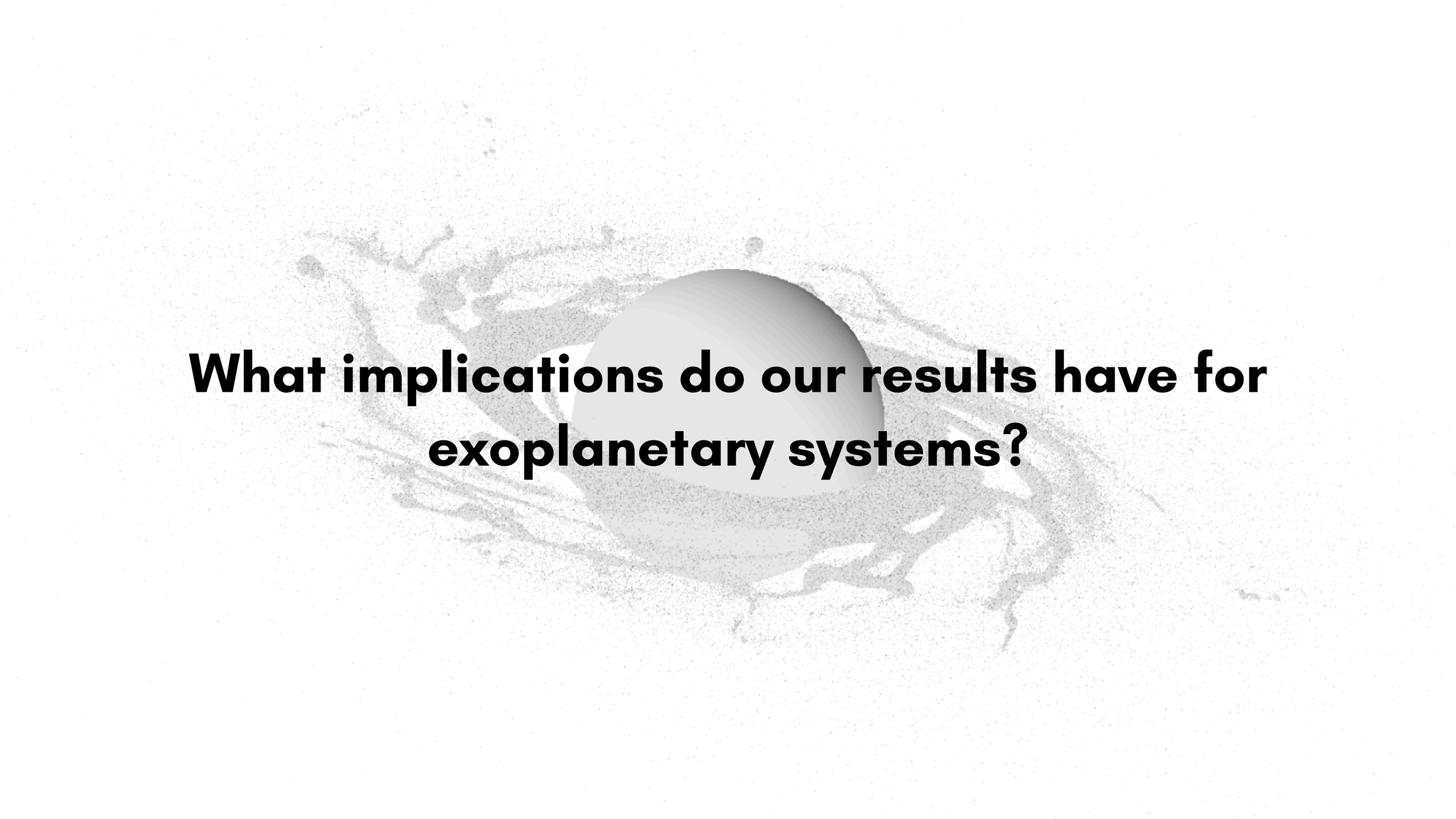
IMPLiCATiONS FOR KNOWN SATELLiTE SYSTEMS



Formation around terrestrial planets and super-Earths (non-tidal regimes) follows an empirical law:

$$M_{\mathrm{l,sat}} = (0.20 \pm 0.16) M_{\mathrm{disk}}^{0.98 \pm 0.16}$$

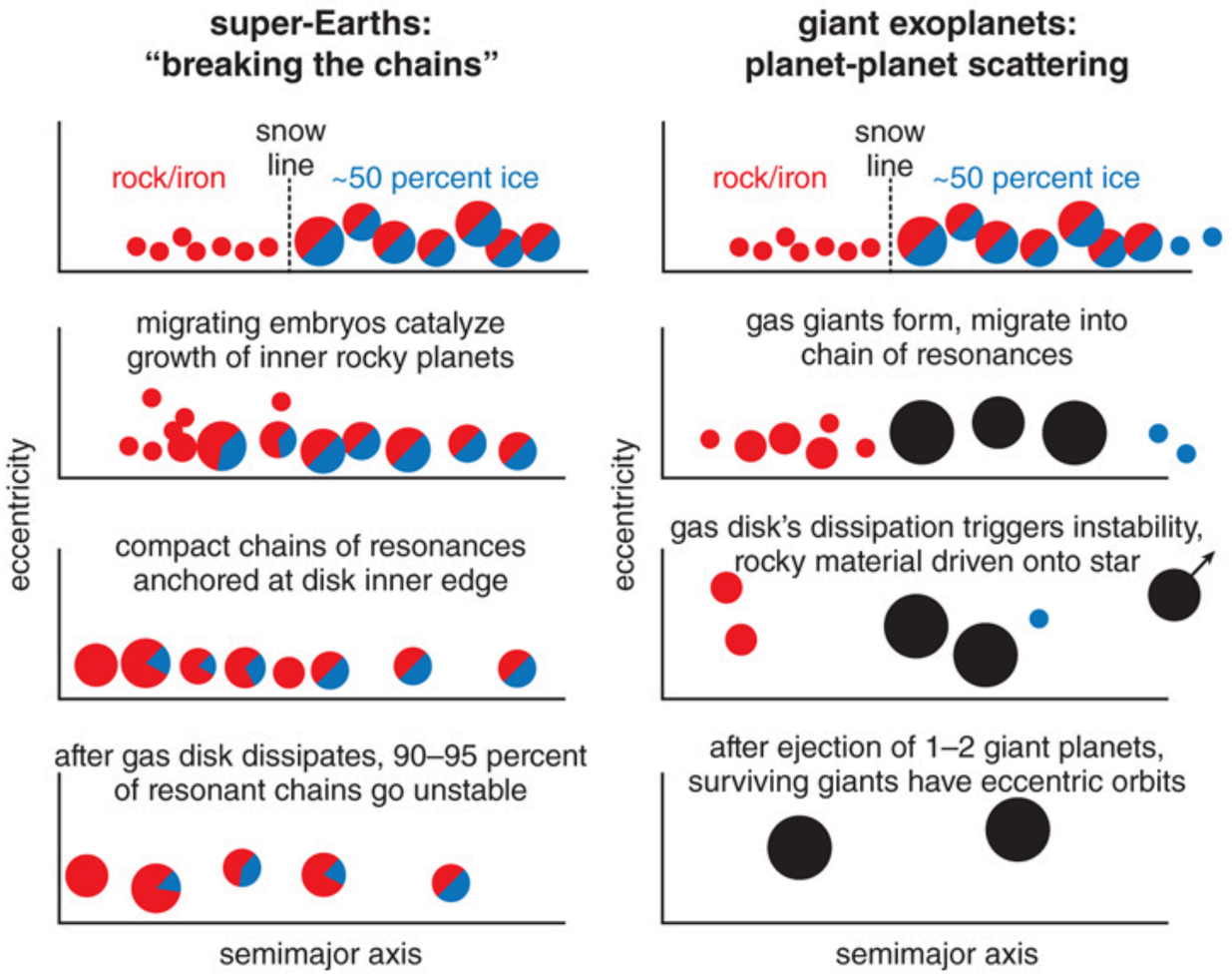
- Proto-lunar disk: 0.03–0.1 Mc
  - Consistent with SPH simulations (Canup, 2012; Ćuk & Stewart, 2012; Meier et al., 2024)
- Charon's accretion would require M_disk ⩾ 0.3 Mc
  - Inconsistent with SPH simulations (Canup, 2011)
  - Charon likely formed as a direct fragment of the impact
  - The remaining satellites would form from a disk with Mdisk of about 10^{-4} Mc (pyramidal regime)



What implications do our results have for exoplanetary systems?

Two main formation pathways for terrestrial planets and super-Earths

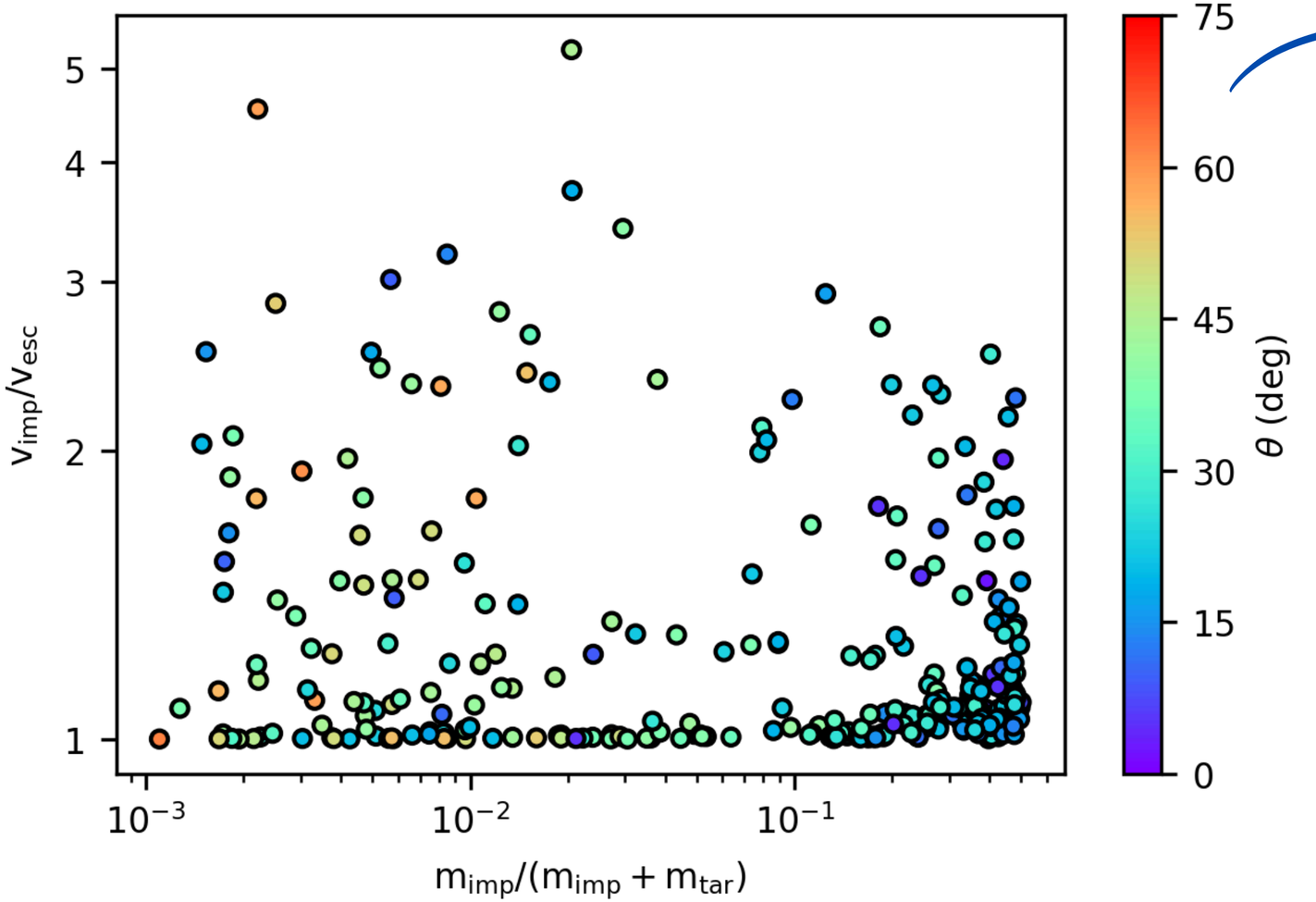
- The formation of a giant planet may or may not isolate the terrestrial planet-forming region, affecting their masses and composition (Lambrechts et al., 2019; Izidoro et al., 2019; Bitsch et al., 2019).



In any case, the formation of these objects primarily occurs through impacts between protoplanets.

Estimating exomoon masses

- We isolate the final impact responsible for the formation of terrestrial planets and super-Earths in the simulations of Lambrechts et al. (2019), Izidoro et al. (2019), and Bitsch et al. (2019):



Canup (2005): Disk mass as a function of impact parameters

$$f(\gamma) = \gamma(1-\gamma)[\gamma^{1/3} + (1-\gamma)^{1/3}]^{1/2}$$

$$J_{\rm col} = \sqrt{2} f(\gamma) \sin\theta \left(\frac{v_{\rm imp}}{v_{\rm esc}}\right)$$

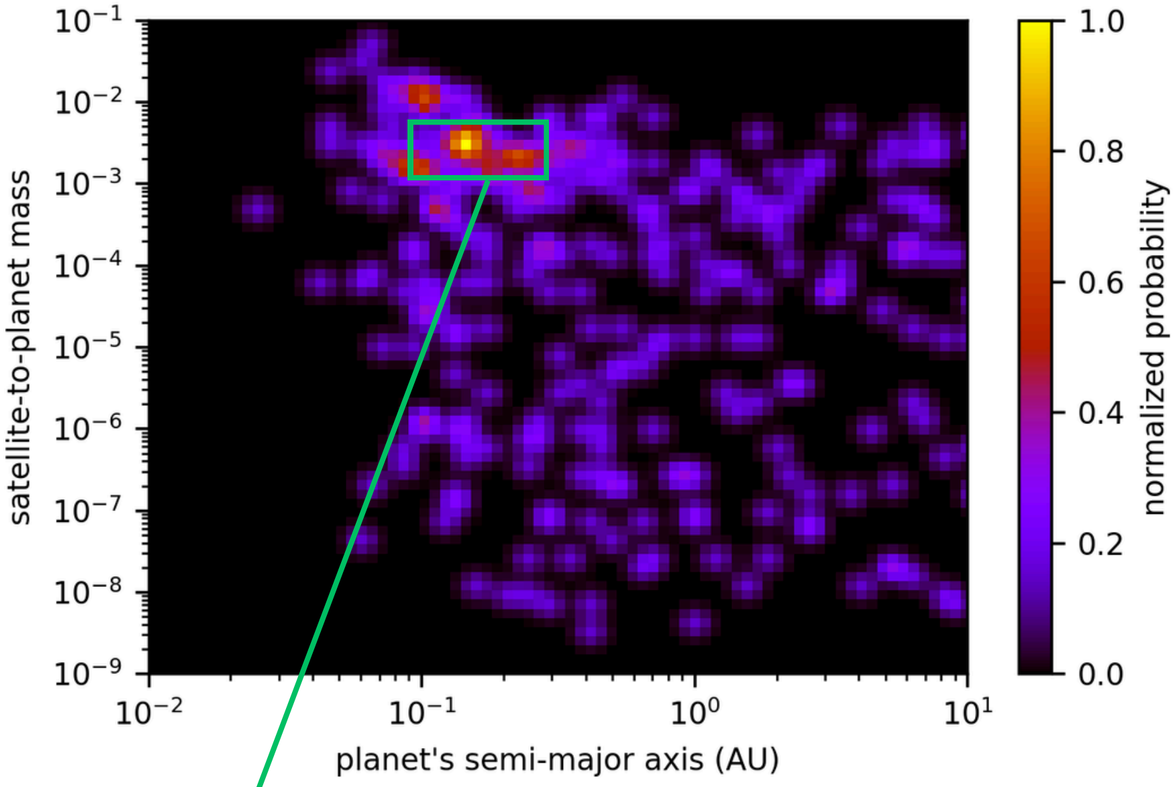
$$M_{\rm o}/M_{\rm T} \approx 2.15 J_{\rm col}^{2.79}$$

Madeira et al. (2025b): largest satellite mass

$$M_{\rm l,sat} = (0.20 \pm 0.16) M_{\rm disk}^{0.98\pm0.16}$$

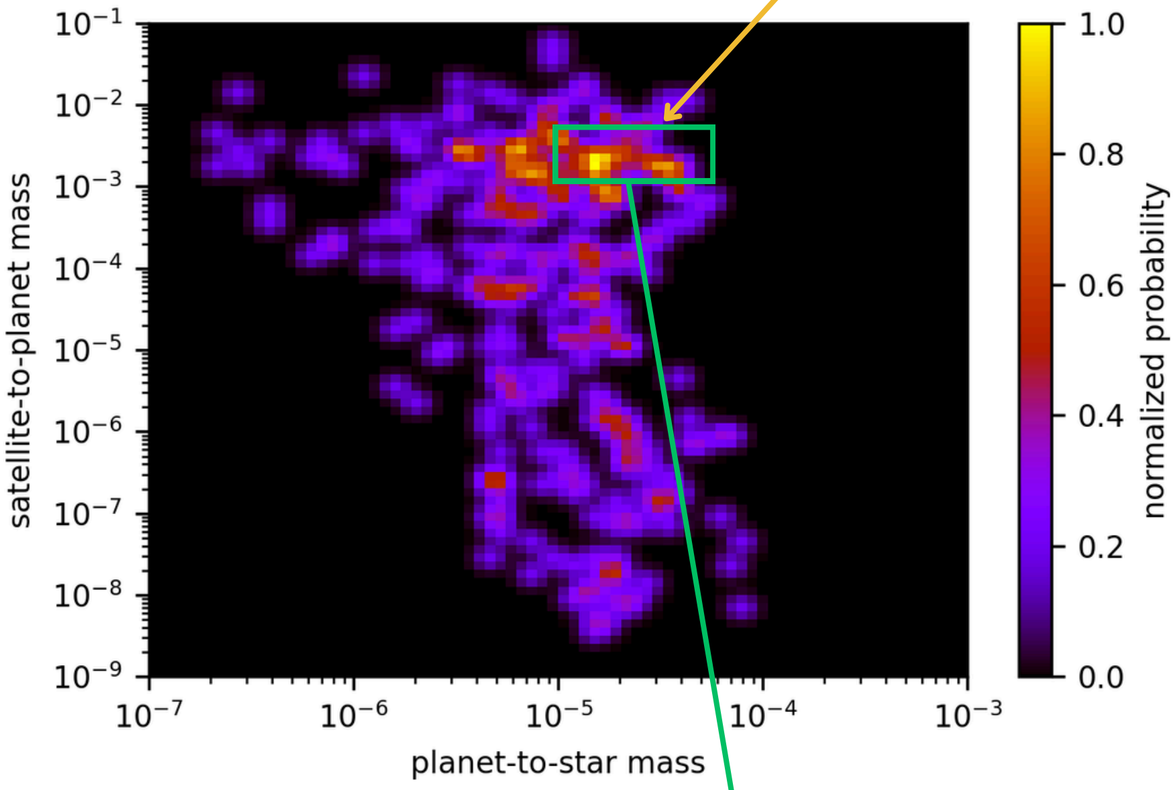
Estimating exomoon masses

- Satellite occurrence probability:



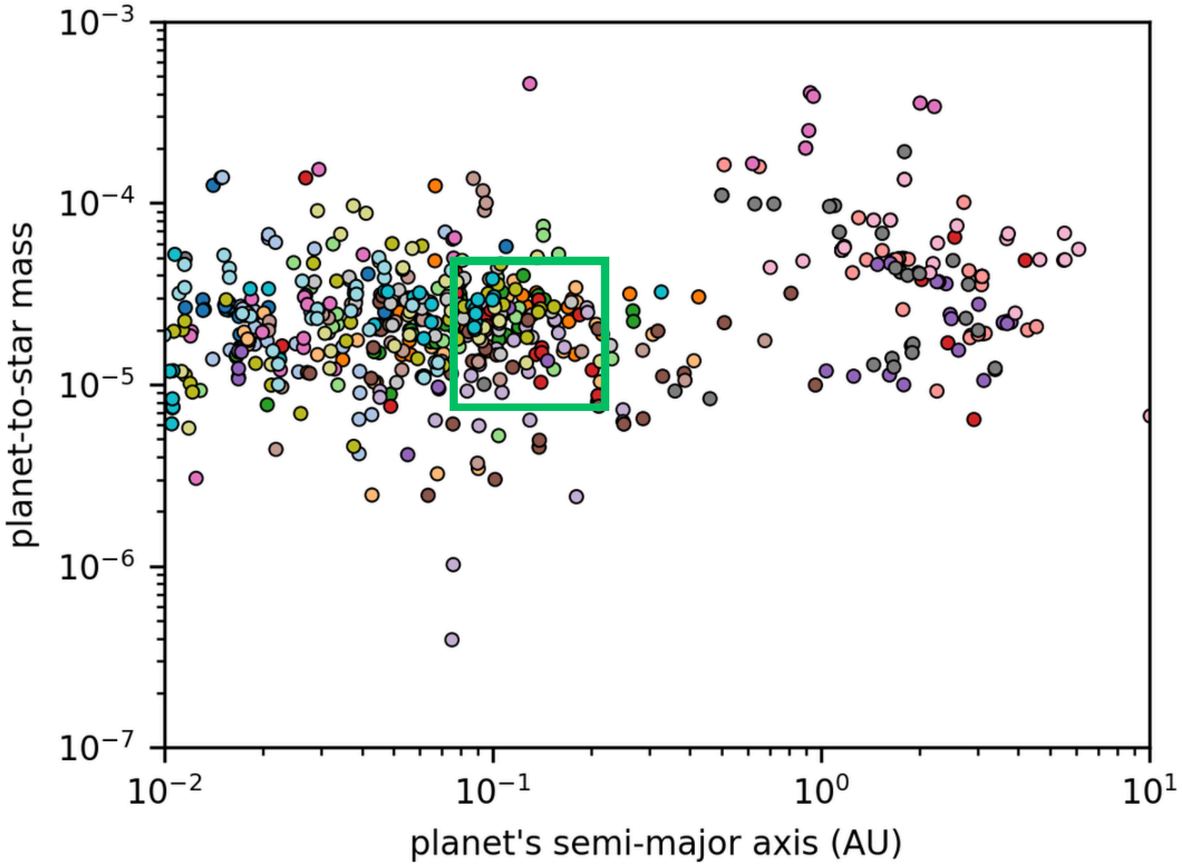
Higher probability for planets at 0.08–0.3 AU

Satellite masses:
10^{-3}–10^{-2} Mc



Higher probability for planets with masses of 7×10^{-6}–7×10^{-5} times the stellar mass

Estimating exomoon masses



Confirmed terrestrial planets and super-Earths (Exoplanet Archive)

According to the models considered, super-Earths most likely to host large satellites are those with:

- Masses: 7×10^{-6}–7×10^{-5} times the stellar mass
- Semi-major axis: 0.08–0.3 AU

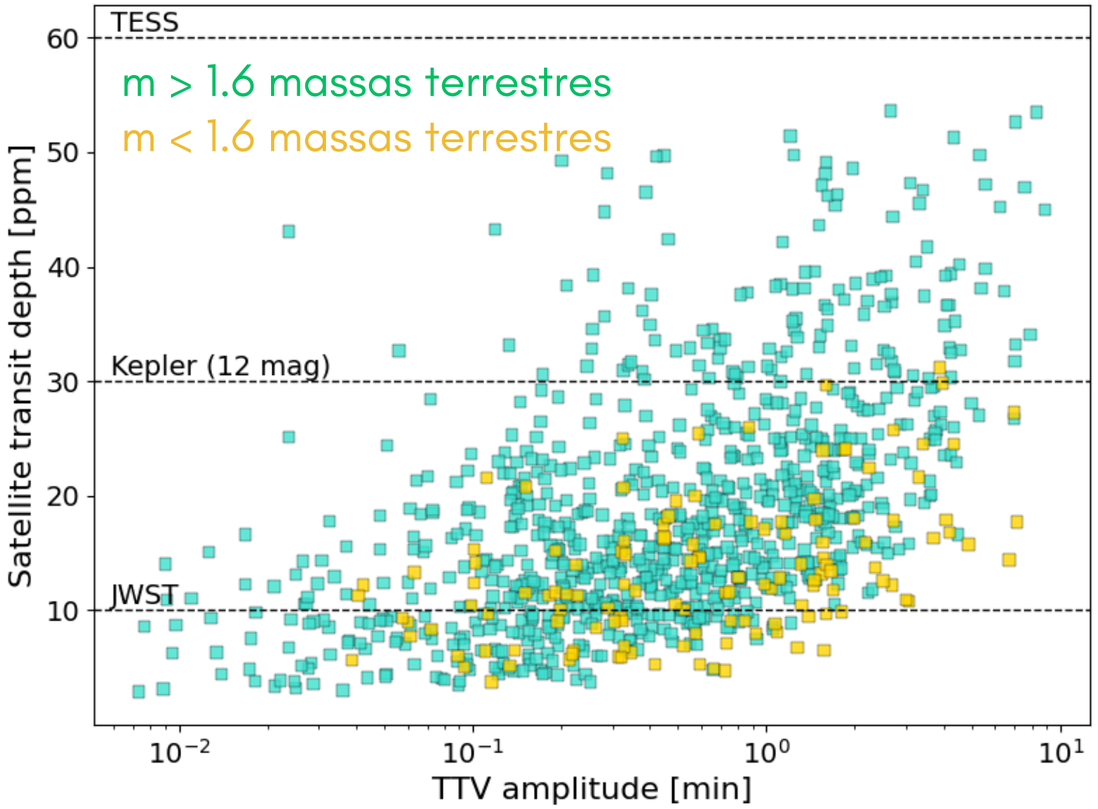
Expected satellite masses:

- 10^{-3}–10^{-2} times the planetary mass

Would these exomoons be observable?

Consider a satellite–planet–star system:

- Planet mass = 0.5–10 Earth masses
- Semi-major axis = 0.01–0.5 AU
- Satellite masses = 10^{-3}–10^{-2} times the planetary mass



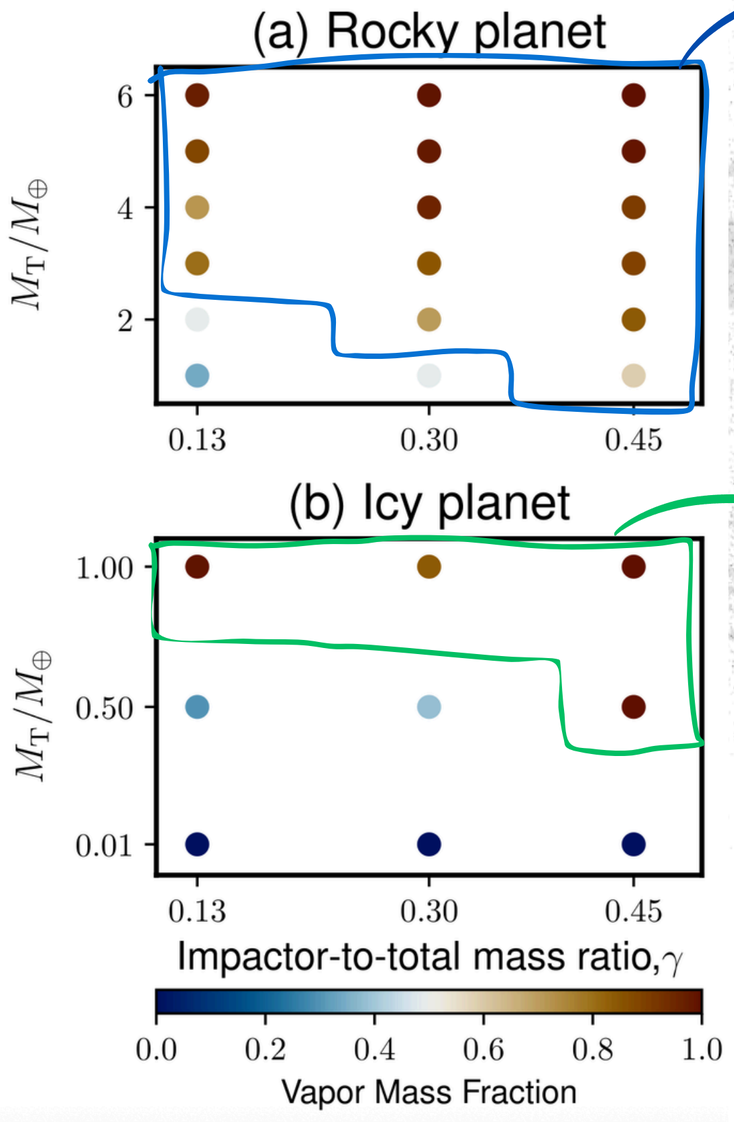
Pinheiro et al. (2026, MNRAS)

- Typical satellites around planets with masses > 1.6 Earth masses would be readily detectable via transits with James Webb Space Telescope and Kepler Space Telescope

- **For planets with masses < 1.6 Earth masses, only satellites as massive as the Moon would be detectable via transits with JWST**

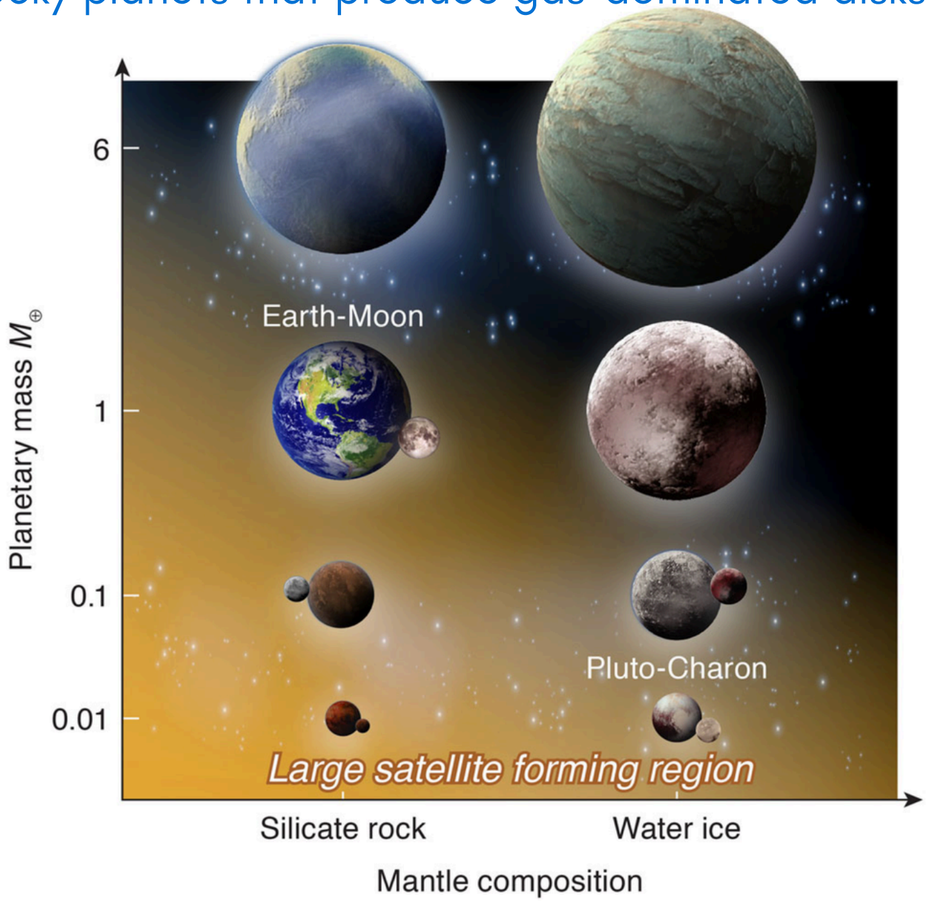
So why have no exomoons been detected so far?

- The answer may lie in Nakajima et al. (2022)



Impacts involving rocky planets that produce gas-dominated disks

Impacts involving icy planets that produce gas-dominated disks



3D simulations of two-phase disks (solid + gas) are needed to better understand satellite formation around super-Earths (a project for a future graduate student)

QUESTiONS?